# Chapter 2
# Semiconductor Properties

## 2.1 Introduction

Research on semiconductors has a long history [Lar54, Smi59]. Phenomenologically they are defined as substances whose electrical resistivity covers a wide range, about $10^{-4}$ to $10^{9}$ Ωcm, between that of metals and insulators and which at high temperatures decreases with increasing temperature. Other characteristics are light sensitivity, rectifying effects and, most typical, an extreme dependency of the properties even on minute impurities. After reaching a basic understanding of their physical nature in the 1930s and 1940s, semiconductors are defined now often by the energy band model and impurity levels leading to the observed phenomena: Semiconductors are solids whose conduction band is separated from the valence band by an energy gap $E_g$ and at sufficiently low temperatures is completely empty, whereas all states of the valence band are occupied. Most important for application in devices, however, is that the conductivity can be controlled over a wide temperature range by impurities, and that there are two types of impurities, donors which release electrons causing n-type conductivity and acceptors which provide positive carriers, the holes, leading to p-type conductivity. This allows the fabrication of pn-junctions.

Semiconductors for power devices must have a sufficiently large energy gap or 'band gap' to insure that the intrinsic carrier concentration, present also without doping, stays below the doping concentration of the weakest doped region up to a sufficient operation temperature, for example 450 K. This is the precondition that the doping structure remains effective. A sufficient band gap is also advantageous because the critical field strength, up to which the material can withstand electric fields without breakdown, increases with band gap. The critical field is a key parameter for the design and function of power devices. Unnecessary large energy gaps, on the other hand, can prove disadvantageous because the ionization energy of the impurities becomes larger with increasing band gap and hence they release only an unfavorably low number of carriers at room temperature. Also built-in and threshold voltages get larger with wider band gap.

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Other properties of the semiconductor are also important. An essential demand is first that the mobilities of free electrons and holes are sufficiently high. Furthermore most power devices based on high carrier injection require a semiconductor with an indirect bandgap such as silicon (see Sect. 2.4). Only these semiconductors allow a sufficient control of the excess carrier lifetime, which determines dynamical and stationary characteristics. Of course also the properties of impurities needed for the pn-structure and adjusting the lifetime are absolutely essential. As a significant chemical semiconductor property we mention the stable native oxide of silicon, which is a prerequisite for almost the whole modern semiconductor technology. A detailed description of the relevant semiconductor properties follows in the present Chap. 2, but their impact on device characteristics and technology will come up at many points in the book.

Semiconductors for power devices are always mono-crystalline, of single crystal form, because, first, only single crystals guarantee a homogeneous space charge under blocking bias and as few as possible energy levels in the band gap. High blocking voltages and low leakage currents are possible only under these conditions. Second, carrier mobilities in single crystal semiconductors are much higher than in polycrystalline. This allows correspondingly higher current densities and the use of smaller devices.

The advantages obtained passing from poly- to mono-crystalline material can be seen looking at the first commercial semiconductor power rectifiers made of poly-crystalline selenium. These rectifiers were produced from about 1940 until 1970 [Spe58, Pog64]. With current densities of at best 1 A/cm$^2$ and blocking voltages up to 40 V per cell, packages with relatively large volume were necessary to handle the given currents und voltages. The properties were to a good part due to the poly-crystalline state, although partly also to the semiconductor Se itself. Silicon diodes replacing Se rectifiers (and Ge diodes) from the end of the 1950s allow current densities and blocking voltages more than two orders of magnitude higher.

The era of modern semiconductors started at the end of the 1940s with the advent of hyper-pure single crystal germanium together with a breakthrough in understanding pn-junctions and the invention of the transistor. Ge was replaced by silicon in the 1950s, and nowadays it has no importance for power devices. Its main drawback is the relative small band gap of only 0.67 eV (at 300 K) which results in a low allowed operating temperature ($\approx$70 °C). Si is by far the most commonly used semiconductor also for power devices, and one can read more about it in this book. The band gap of 1.1 eV and other properties of Si are very suitable for most applications. Si belongs to the group IV of the periodic system, following carbon, C, and preceding Ge. A compound semiconductor consisting of the group III element gallium and the group V element arsenic is gallium-arsenide, GaAs. This semiconductor has gained significance for microwave devices, since it allows high switching frequencies. It has a bandgap of 1.4 eV and a very high electron mobility (five times that of Si). In the field of power devices, the high electron mobility renders it suitable for high voltage Schottky diodes. Schottky diodes of GaAs for 300 V and higher are available on the market.

A compound semiconductor, which after much research and development efforts in the last decades has gained now high significance for power devices, is silicon carbide, SiC [Fri06]. It has a large band gap ranging from 2.3 to 3.3 eV depending on the crystal modification (polytype, see the next section). Consequently, the maximum operating temperature up to which the intrinsic carrier concentration is small enough, and the critical field strength at which breakdown occurs are much higher than for Si. Whereas the higher possible temperature could not be utilized much till now because metallurgical properties of contacts and packaging limits the operating temperature, extensive use was made of the benefits of the high critical field, which are a much higher allowed doping concentration and smaller width of the base region required for a given blocking voltage. This means that fast Schottky diodes and other unipolar devices for high voltages can be fabricated, and pn-devices with extremely high blocking voltage (possibly up to 20 kV) and low power losses are possible. The main advantage of SiC devices is that they allow higher operation frequencies. A difference to Si technology is that diffusion of impurities cannot be used as a doping method because the diffusion constants in SiC are too small. Although the technology of SiC is less versatile and developed and more expensive than that of Si, the potential of SiC devices in power electronics is high. Not only Schottky diodes and MOSFETs for currents up to about 50 A and voltages up to 1700 V are on the market but also "full SiC" power modules for hundreds of amperes and voltages up to 1200 V and more. A phase of intensive testing is going on with the hope of widespread applications. To take advantage of the very small switching times, gate circuits and electronic environment have to be fundamentally readapted to keep inductive voltage peaks and high-frequency oscillations below a dangerous level. We will deal with SiC devices at the proper points of the book.

In recent years, intensive investigations have been performed on microwave and power devices on the base of GaN [Ued05]. This is a semiconductor which with regard to the bandgap of 3.4 eV and a correspondingly high critical field is similar to 4H-SiC. Although the technology has profited greatly from the developments in the LED area, the fabrication of large single crystals and wafers of GaN offers still problems. GaN-devices demonstrated till now are lateral devices using GaN films grown epitaxially on substrates of silicon, 4H-SiC or saphire ($Al_2O_3$). 8-inch GaN-on-Si wafers can meanwhile be produced. A basic element of GaN devices is a highly conducting electron layer ('two-dimensional electron gas', 2DEG) induced by the high electric polarization of GaN and AlGaN which latter is deposited as a thin layer on the GaN. Owing to the high critical field the dimensions can be chosen small, so a small on-resistance can be obtained per area. Field effect transistors with a blocking voltage up to 1.8 kV and a current of more than 100 A have been demonstrated [Ike08]. The main interests are at mid-range voltages (<1 kV) for high switching frequency applications. Special devices of the 600 V class are on the market [Mor14, Ish15, Hon15]. Compared with SiC the developments are still in an earlier stage.

A further subject of investigations in view of a possible future material for power devices has been the diamond modification of carbon. Diamond has an energy gap

of 5.5 eV and in pure form is an insulator. Its critical field is extremely high (10–20 MV/cm). If it contains impurities like boron or phosphorus it behaves like a semiconductor whose carriers have higher mobilities than in silicon. A handicap of diamond is the high ionization energy of the dopants, especially of donors. Hence the impurities are ionized only to a small extent at room temperature. Concepts to partially overcome this drawback have been proposed and samples of high-voltage Schottky diodes have been demonstrated [Twi04, Ras08]. The employment of diamond devices in power electronics is expected at best in the longer term.

## 2.2 Crystal Structure

The atoms in a crystal are arranged in a three-dimensional lattice at lattice points around which they can vibrate. The lattice is to a great extent determined by the type of bonding between the atoms. In all the named semiconductors (apart from Se) the bonds in all directions are (widely) covalent, i.e. formed by an electron pair with opposite spins. This is the rule for group IV elements. Because they have four valence electrons in the outer shell, an atom can undergo covalent bonds with four nearest neighbors which, like the central atom, contribute an electron to the respective bond. In this way, a closed shell of eight shared electrons is gained for each atom establishing a very stable configuration. Similar considerations hold for III–V compounds such as GaAs whose constituents *on average* have four valence electrons per atom. Symmetrically arranged the four nearest neighbors are located at the corners of a tetrahedron the midpoint of which is occupied by their common bonding partner. For the element semiconductors, this tetrahedral bonding can be realized only by one lattice structure, the diamond lattice. This is hence the lattice of Ge, Si and the diamond modification of carbon.

Figure 2.1a shows the cubic unit cell of the diamond lattice. It can be described as superposition of two face-centered cubic (fcc) sublattices, which by definition have lattice points at the corners and midpoints of the side faces of the cube. The two sub-lattices are shifted by a quarter of the space diagonal. In the lower left part of the figure, the covalent bonding of an atom on the space diagonal with its four nearest neighbors is marked. This part of the figure is redrawn in Fig. 2.1b to show the tetrahedral bonding geometry more clearly.

GaAs has the same lattice structure, except that now one of the face-centered cubic sub-lattices is occupied by Ga and the other by As atoms. If in Fig. 2.1b an As atom is in the center, the four neighbors at the corners are Ga and vice versa. This so-called zincblende lattice is the structure of most III–V semiconductors [Mad64].

An exception is GaN which crystallizes preferably in the wurtzite structure, a hexagonal lattice called after a second polytype of ZnS appearing besides zincblende. The also appearing zincblende modification of GaN is less stable and not used in devices. In the wurtzite lattice the tetrahedral bonding between the neighbors of both types is realized as well, only the orientation of neighboring tetrahedra

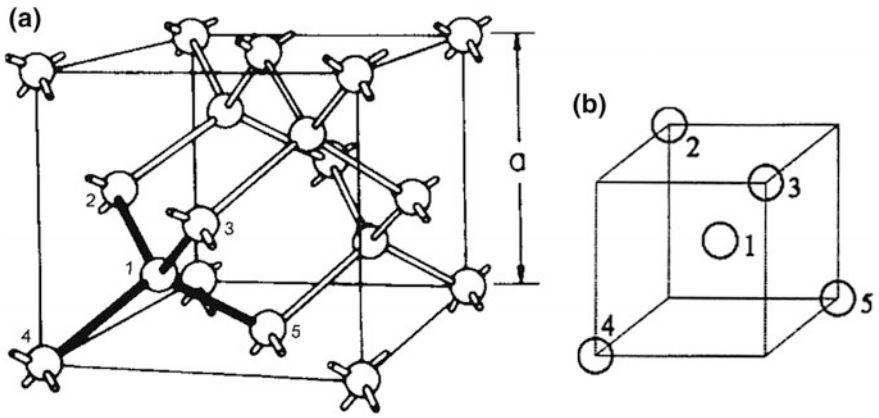



**Fig. 2.1** **a** Cubic unit cell of the diamond lattice. The tetrahedral bonding between an atom 1 on the space diagonal and four nearest neighbors 2–5 is marked. **b** Central atom with four neighbor atoms at the corners of a tetrahedron. The lattice constant for silicon is a = 5.43Å, the distance between the center of Si atoms $d = 3^{1/2}/4\ a = 2.35$ Å (adapted from [Sze81] and [Hag93])

towards each other is different from the zincblende structure. Actually, the tetrahedral bonding system of XY-compounds is compatible with many other lattices. This is shown in fact by SiC which crystallizes in the zincblende structure and likewise in many other polytypes, most of which are hexagonal. In all cases, the arrangement of nearest neighbors is identical, each silicon atom being surrounded by four carbon atoms at the corners of a tetrahedron with the Si in the center, and vice versa. Only the arrangement of more distant atoms varies from polytype to polytype [Mue93]. The atomic distance between nearest neighbors is always 0.189 nm = 1.89 Å, nearly the mean value between the atomic distances in diamond with 1.542 Å and Si with 2.35 Å. For power devices, the hexagonal polytype called 4H-SiC is preferred. The prefix indicates that the structure repeats itself after stacking of four Si-C double planes. In this nomenclature the wurtzite polytype of GaN has a 2H-structure, since the lattice repeats itself after two double planes of Ga-N.

The used crystal structures are compiled in the following list:

| | |
|---|---|
| Ge, Si, diamond-C | Diamond lattice, cubic |
| GaAs | Zincblende lattice, cubic |
| GaN | Wurtzite lattice, 2H-hexagonal |
| 4H-SiC | 4H-hexagonal |

The crystal orientation influences to some extent processing parameters and physical properties. In silicon, the thermal oxidation rate, the epitaxial deposition rate, the density of surface states and the elastic constants depend on orientation, whereas mobilities and diffusion constants due to the cubic lattice are (isotropic) scalars. To describe crystal planes and directions one uses the Miller indices. These

are defined via the reciprocals of the intercepts of the considered plane with the crystallographic axes, expressing their ratios by a triple of smallest integers. For example, the six faces of a cube intersect two of the axes in the point $\infty$, hence two of the Miller indices are 0, and the planes are denoted as (1 0 0), (0 1 0), (0 −1 0) and so on. The family of these planes is denoted as {100} planes. The crystal planes which have equal intercepts on the axes are the {111} planes. In silicon technology crystal wafers (slices) with a {100} or a {111} surface are used.

## 2.3 Energy Gap and Intrinsic Concentration

The energy gap results because all the valence electrons of the semiconductor atoms are required for the complete covalent bonds and because a certain amount of energy is necessary for breaking electrons out of the bonds so that they can move freely and hence conduct current. This is expressed by the band diagram shown in Fig. 2.2, where the energy of the electrons is plotted versus a space coordinate $x$.

The valence band contains the states for the electrons in the bonds, and the conduction band represents the states of the electrons which are free for conduction. Between the top of the valence band, $E_V$, and the bottom of the conduction band, $E_C$, no energy levels are present, $E_g = E_C - E_V$ is the energy gap. Here we refer to an "intrinsic" semiconductor whose properties are not influenced by impurities but are due to the semiconductor itself. At least the energy $E_g$ is necessary to break an electron out of the bonds and create a conduction electron. At zero absolute temperature, this energy is not available for the valence electrons (external activation excluded), and since no carriers are present, the semiconductor behaves like an insulator. At $T > 0$ a certain number of valence electrons is elevated into the conduction band. By this process, however, not only conduction electrons are created, but also an equal number of empty electron places remain in the valence band and these voids, called "holes", can conduct current likewise. As a missing valence electron the hole has the opposite, i.e. a positive charge. Because in an electric field the empty place is filled repeatedly by a neighboring valence electron, the hole travels in opposite direction to the electrons. This is similar to a bubble in water moving against gravity. It follows, that one can ascribe also an energy to the

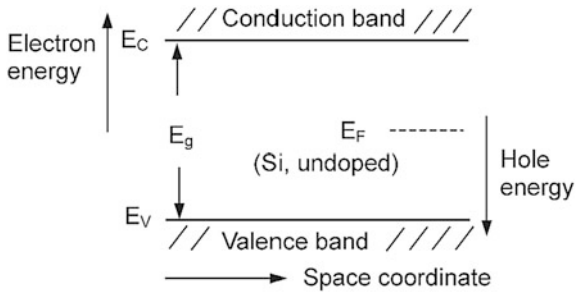



**Fig. 2.2** Energy band model

holes which is directed downward in Fig. 2.2 (see the calculation below). So far, the picture of a (classical) vacancy at a point $x$ in the otherwise full band of valence electrons is successful. With this understanding the hole is an auxiliary quantity for a simpler description of the motion of all involved valence electrons which are the actual carriers. However there are other decisive experiments, particularly measurements of the Hall effect, which contradict this classical picture and show the holes as positive charge carriers on their own. This is confirmed by quantum theory according to which the holes are independent stable (quasi-) particles on the same grounds as conduction electrons. A detailed discussion of these features will be given in the next paragraph.

We calculate now the intrinsic concentration of carriers in a pure semiconductor and introduce simultaneously some more general relationships for electrons and holes which are applicable also when impurities are present. Since the thermal generation of carriers is counteracted by recombination annihilating them, this leads to a thermal equilibrium described by statistical physics. The occupation probability of states with energy E, defined as the number of electrons $n_E$ per number of states with that energy, $N_E$, is given by the Fermi distribution

$$\frac{n_E}{N_E} = \frac{1}{1 + e^{\frac{E-E_F}{kT}}} \tag{2.1}$$

In this equation $k$ denotes the Boltzmann constant, $T$ the absolute temperature and $E_F$ the Fermi level, which in statistical thermodynamics is known under the name "chemical potential". Fermi energy $E_F$ is a constant of the system determined so that the sum over $n_E$ returns the total electron concentration. Equation (2.1) is a consequence of the quantum mechanical principle, that a state cannot be occupied by more than one electron. As shown by the equation, the states with energy smaller than $E_F$ are mostly occupied by an electron, while the states with $E > E_F$ are mostly empty. The number of occupied divided by the number of unoccupied states of energy E, called the occupation degree, is

$$\frac{n_E}{N_E - n_E} = e^{-(E-E_F)/kT} \tag{2.2}$$

In the intrinsic case and often also in doped semiconductors (see next chapter), the occupation probability is small, $n_E << N_E$. In this so-called non-degenerate case, (2.2) turns into the classical Boltzmann or Maxwell-Boltzmann distribution:

$$\frac{n_E}{N_E} = e^{-(E-E_F)/kT} \tag{2.3}$$

Since primarily only states near the bottom of the conduction band are occupied and primarily only states near the top of the valence band are empty, we assume at first that the states in the bands are concentrated at the edges and their number in integrated form is given by effective densities of states (number per volume) $N_C$, $N_V$.

Taking (2.3) at the energy $E_C$ one obtains for the concentration of conduction electrons in thermal equilibrium:

$$n = N_C \cdot e^{-(E_C - E_F)/kT} \tag{2.4}$$

Using (2.2) at $E = E_V$ and considering that the density of unoccupied states in the valence band is identical with the hole concentration p, whereas the density of occupied states is $N_V - p \approx N_V$, the hole concentration in thermal equilibrium is obtained as follows:

$$p = N_V \cdot e^{-(E_F - E_V)/kT} \tag{2.5}$$

This shows that the holes behave statistically as particles, with energy scale inverted compared with electrons. Multiplication of (2.4) and (2.5) using $E_C - E_V = E_g$ yields:

$$n \cdot p = n_i^2 = N_C N_V \cdot e^{-E_g/kT} \tag{2.6}$$

where $n_i = n = p$ is the intrinsic concentration. Equation (2.6) is the mass law equation of the reaction $(0) \leftrightarrow n + p$, describing generation and, inversely, the recombination of an electron-hole pair.

Since the condition $n = p = n_i$ has not been used in the derivation, the intrinsic conduction represents only a special case of (2.4) (2.5) and (2.6), actually they are applicable also to doped semiconductors where $n \neq p$. Doped semiconductors will be treated in detail in one of the next sections. Only the Boltzmann distribution (2.3) is assumed for thermal equilibrium, meaning that the doping is not too high (case of non-degeneracy). As shown by (2.6), the np product is a constant independent of the Fermi level, but dependent on the bandgap and temperature.

The Fermi level for the intrinsic case is obtained from (2.4) and (2.5) setting $n = p$:

$$E_i = \frac{Ev + Ec}{2} - \frac{kT}{2} \cdot \ln \frac{Nc}{Nv} \tag{2.7}$$

Because of similar values of the densities of states $N_C$, $N_V$, the Fermi level in intrinsic semiconductors is located close to the middle of the bandgap.

In spite of the simplifying assumption on the distribution of the states in the bands, (2.4), (2.5) and (2.6) are applicable to the actual situation. To take into account that with increasing T more states above respectively, below the band edges are occupied, this results only in a temperature dependency of the effective densities of states $N_C$ and $N_V$. The density of states $N_E$ increases with distance ΔE from the edges as $\sqrt{\Delta E}$. Multiplying this with the Boltzmann factor of Eq. (2.3) and integrating, one obtains again (2.4) and (2.5), where $N_C$, $N_V$ now are proportional to $T^{3/2}$ [Sze02]. Considering that also the band parameters themselves vary a little with $T$, one obtains for Si [Gre90]:

$$N_C = 2.86 \times 10^{19} \left(\frac{T}{300}\right)^{1.58} /\mathrm{cm}^3$$
$$N_V = 3.10 \times 10^{19} \left(\frac{T}{300}\right)^{1.85} /\mathrm{cm}^3 \tag{2.8}$$

These numbers are large compared with the doping concentrations in most cases, as will be seen later. Compared with the number of Si atoms per $\mathrm{cm}^{-3}$, $5.0 \times 10^{22}$, they are small.

The bandgap is approximately a constant. However, seriously considered, it decreases slightly with temperature. This can be expressed for Si and other semiconductors in the form [Var67]:

$$E_g(T) = E_g(0) - \frac{\alpha \cdot T^2}{(T + \beta)} \tag{2.9}$$

The bandgap parameters of this equation together with the effective densities of states are compiled for Si, GaAs, 4H-SiC and GaN in Table 2.1 [Thr75, Gre90, Lev01, Mon74].

The intrinsic carrier densities calculated with these data for Si, GaAs, 4H-SiC are shown in Fig. 2.3 as functions of temperature. Also germanium is included. From Si to SiC the intrinsic concentration decreases extremely for a given temperature due to the exponential dependence on energy gap. Considering a given value of $n_i$, the absolute temperature at which this value is adopted is proportional to $E_g$, if the temperature-dependence of the pre-exponential factor in (2.6) is neglected.

The temperature at which $n_i$ reaches a value comparable with the impurity concentration of the lowest doped region in a device, defines a limit, above which the pn-structure begins to be leveled out and loses its normal function. If $n_i$ is dominating, its exponential increase with temperature and the corresponding decrease of resistance, together with thermal feedback, can lead to current constriction and destruction of the device. In a Si device with 1000 V blocking voltage, a doping in the range of $10^{14}$ $\mathrm{cm}^{-3}$ is necessary for the base region to sustain the voltage. To meet the condition $n_i < 10^{14}$ $\mathrm{cm}^{-3}$, the temperature must remain below about 190 °C, as is shown in Fig. 2.3. With 4H-SiC, temperatures of more than

**Table 2.1** Bandgap parameters and effective density-of-states of some semiconductors

| | Si | GaAs | 4H-SiC | GaN |
|---|---|---|---|---|
| $E_g(0)$ (eV) | 1.170 | 1.519 | 3.263 | 3.47 |
| $\alpha * 10^4$ (eV/K) | 4.73 | 5.405 | 6.5 | 7.7 |
| $\beta$ (K) | 636 | 204 | 1300 | 600 |
| $E_g(300)$ (eV) | 1.124 | 1.422 | 3.23 | 3.39 |
| $N_C(300)$ ($\mathrm{cm}^{-3}$) | $2.86 \times 10^{19}$ | $4.7 \times 10^{17}$ | $1.69 \times 10^{19}$ | $2.2 \times 10^{18}$ |
| $N_V(300)$ ($\mathrm{cm}^{-3}$) | $3.10 \times 10^{19}$ | $7.0 \times 10^{18}$ | $2.49 \times 10^{19}$ | $4.6 \times 10^{19}$ |

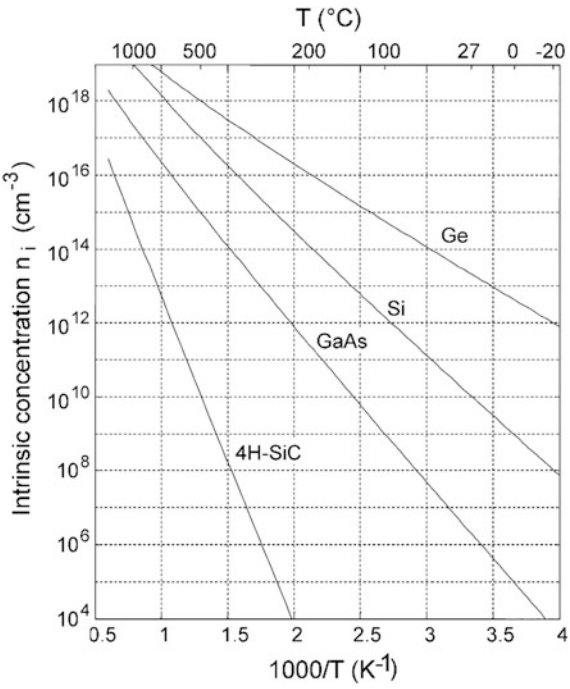



**Fig. 2.3** Intrinsic carrier density for Ge, Si, GaAs and 4H-SiC as a function of temperature

800 °C would be allowed for 1000 V devices from this requirement. In practice, interconnect and packaging materials, however, set a much lower temperature limit.

Equation (2.6) shows why semiconductors with a wide bandgap, if nearly intrinsic, behave like insulators. In 4H-SiC, the intrinsic concentration as given by above data is even at 400 K only 0.3 $cm^{-3}$, corresponding to a resistivity of $\approx 2 \times 10^{16}$ Ωcm. Although the practically attainable resistivity is several orders of magnitude smaller, SiC can be used for insulating layers with high thermal conductance.

For later use in the case of doped semiconductors we note an alternative form of (2.4) and (2.5) which is obtained using the intrinsic concentration and intrinsic Fermi level to eliminate the effective densities of states and energies of band edges. Dividing (2.4) by the specialized form $n_i = N_C \exp(-(E_C - E_i)/kT)$ it converts to:

$$n = n_i \cdot e^{\frac{E_F - E_i}{k \cdot T}} \tag{2.10}$$

The hole concentration in thermal equilibrium can be written as follows:

$$p = n_i \cdot e^{\frac{E_i - E_F}{k \cdot T}} \tag{2.11}$$

## 2.4 Energy Band Structure and Particle Properties of Carriers

Besides the E(x) band diagram of Fig. 2.2, there is the more detailed energy band representation in k-space, $E(\vec{k})$, which allows further insight into fundamental semiconductor properties. Here the electron energy is plotted versus the wave vector $\vec{k}$ of a wave packet, which solves the quantum mechanical Schrödinger equation for an electron in the lattice-periodic potential of the crystal. Of main interest are the valence $E(\vec{k})$ band with highest maximum and the conduction $E(\vec{k})$ band with lowest minimum. In Fig. 2.4 these bands are shown for Si and GaAs for specified directions in $\vec{k}$-space.

The energy difference between the absolute minimum of the conduction band and the maximum of the valence band is the band gap $E_g$ shown in Fig. 2.2. The maximum of the valence band is nearly always at $\vec{k} = 0$. In GaAs, also the minimum of the conduction band is located at this position. A semiconductor of this kind is called a direct semiconductor. In Si, the conduction band has a minimum far away from $\vec{k} = 0$. Semiconductors of this type are called indirect. Since in Si a minimum lies in each of the {100} directions, there are six minima in a unit cell.

Whether a semiconductor has a direct or indirect band gap is decisive for the probability of transitions between the bands. This determines the suitability for

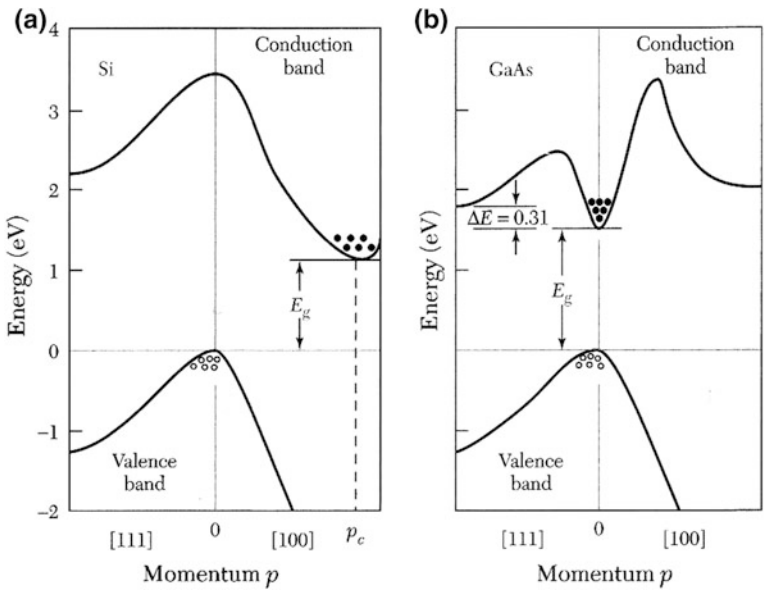



**Fig. 2.4** Energy band E(k) for Si (indirect semiconductor) and GaAs (direct semiconductor). Figure from Sze, [Sze02], reproduced with permission of John Wiley & Sons Inc.

optical devices, but has some significance also for power devices. The influence on transition probability follows because the crystal momentum $\vec{p} = \hbar\vec{k}$ ($h = \hbar/2\pi, = h/2\pi, h$ is Planck's constant) has the property of a momentum of the electron regarding its reaction on external forces. Internal forces arising from the periodic potential in the crystal are taken into account in another way (see below). An external force $\vec{F}$, owing to an electric or magnetic field, causes an acceleration which, as in Newton's second law of mechanics, is given by $d\vec{p}/dt = \vec{F}$. In transitions between bands the change in crystal momentum $\vec{p}$ has to be counted in the law of conservation of momentum. The recombination of an electron at the bottom of the conduction band with a hole at $\vec{k} = 0$ occurs under emission of a photon, which receives nearly the whole energy released, but has negligible momentum. In indirect semiconductors the recombination can occur only if the crystal momentum of the conduction electron can be transferred to a quantum of lattice vibrations, a phonon. Hence, compared with a direct semiconductor such as GaAs, where the recombination occurs without participation of a phonon, the radiative band-to-band recombination in indirect semiconductors has a much lower probability. Therefore, only direct semiconductors are used for LEDs and lasers. The lifetime associated with the radiative band-to-band recombination represents an upper limit of the lifetime of minority carriers. For GaAs, a radiative minority carrier lifetime $\tau = 1/(B \cdot N)$ (B is the recombination constant) of 6 µs is estimated for a doping concentration of $N = 1 \times 10^{15}$ cm$^{-3}$; for $10^{17}$ cm$^{-3}$ this means $\tau = 60$ ns [Atk85]. This is sufficient to allow satisfactory operation of pin diodes for a medium voltage range, but it is detrimental for bipolar transistors and thyristors with commonly used doping structures. In Si the radiative recombination constant B is four orders of magnitude smaller [Sco74]. Hence very high lifetime values are possible in Si. The recombination radiation in Si, having a wavelength $\lambda \cong hc/E_g = 1.1$ µm, is often used as a tool to investigate and test the internal operation of devices. Besides Si also Ge and all polytypes of SiC are indirect semiconductors. Some of the III–V compound semiconductors are of the direct type like GaAs and GaN, while others are of the indirect type like GaP.

The $E(\vec{k})$ bands are also basic for the behavior of electrons and holes as charge carriers. We will outline this here shortly, referring for a more complete discussion to the books of Moll [Mol64] and Spenke [Spe58]. If an external force is applied, the electrons near the minimum of the conduction band are accelerated. However, the increase in kinetic energy is only relatively small, because the acceleration is stopped after a short relaxation time owing to the non-ideality of the crystal, i.e. by scattering by phonons and impurities. Thus the wave vector and kinetic energy are reduced nearly to their initial values (on the statistical average) and remain in the band not far away from the minimum. Hence the kinetic energy, $E_{n,kin} = E - E_C$, can be expressed using Taylor's expansion as

$$E_{n,kin} = \frac{1}{2}\frac{d^2E}{dk^2} \cdot (\vec{k} - \vec{k}_m)^2 \tag{2.12}$$

where $\vec{k}_m$ denotes the $\vec{k}$-vector at the band minimum. We assume for simplicity that the $E(\vec{k})$ function depends only on the absolute value $k$ of $\vec{k}$, and not on the orientation. Defining a "particle momentum" as $\vec{p}_n = \hbar\,(\vec{k} - k_m)$, (2.12) turns into

$$E_{n,kin} = \frac{1}{2\hbar^2}\frac{d^2E}{dk^2}\vec{p}_n^2 = \frac{\vec{p}_n^2}{2m_n} \tag{2.13}$$

where we have defined, furthermore,

$$m_n \equiv \hbar^2 / \frac{d^2E}{dk^2} \tag{2.14}$$

$m_n$ has the dimension of a mass and is called the effective mass of the electrons. It has the order of magnitude of electron mass in vacuum, but is not equal to it. The velocity of the electron is $\vec{v}_n = \vec{p}_n/m_n$. Because the momentum vector $\vec{p}_m$ of the conduction band minimum in $\vec{p}_n = \vec{p} - \vec{p}_m$ is constant, the relation between force and acceleration can be written also as $\vec{F} = d\vec{p}_n/dt$. Hence, quantum mechanics leads to the result that conduction electrons in the lattice-periodic potential obey relationships of mechanics; they react on external forces like mass points with a positive effective mass $m_n$ and a negative charge −q. The interaction with the internal field of the periodic potential is taken into account by the effective mass. The effective mass is not a scalar but a tensor, but this is not important for an essential understanding in fact of the model.

Holes behave quite analogously. This is obtained representing a hole by the full valence band minus an electron. The full band does not conduct current because the parts from + $\vec{k}$ and $-\vec{k}$ compensate each other. The contribution of an electron to the current density is $-q\vec{v}_n$, and that of the missing electron, the hole, $+ q\vec{v}_n$. Since the velocity of the hole is identical to that of the missing electron, the equation of motion yields $\dot{\vec{v}} = \vec{F}_n/m_{n,v} = \dot{\vec{v}}_p = \vec{F}_p/m_p$. Here $\vec{F}_n$ denotes the force on the electron, $\vec{F}_p$ the force on the hole, which due to the opposite charge has the opposite sign for electric and magnetic fields, $\vec{F}_p = -\vec{F}_n$. $m_{n,v}$ is the effective mass of a valence electron. Hence we have to define $m_p = -m_{n,v}$ to obtain the classical relation between force and acceleration. Because at the top of the valence band $d^2E/dk^2 < 0$, the effective mass of the valence electron is negative, and that of the hole positive:

$$m_p = -\hbar^2 / \frac{dE}{dk}\,(E = E_V) \tag{2.15}$$

For $E_{kin,p} \equiv E_V - E$ one obtains similarly as for the electrons: $E_{kin,p} = \vec{p}^2/2m_p = m_p\vec{v}_p^2/2$. Hence we have obtained that also holes obey the relationships of mechanics, they behave like mass points with a positive charge q and positive effective mass. Quantum mechanics leads to a particle picture, called quasi-particle model, which is justified on the same basis for holes as for conduction electrons. It has a wide range of applicability and is used throughout device physics.

Quantitatively, the situation is more complicated because the effective mass of electrons in Si and other semiconductors is a tensor and depends strongly on orientation. However, the effective mass entering the mobility and conductivity, the "conductivity effective mass" $m_{n,c}$, is an average over the equivalent minima in k-space, and this averaging yields a scalar in cubic crystals [Smi59]. The conductivity effective mass in Si is $m_{n,c} = 0.27\ m_0$ ($m_0$ free electron mass) [Gre90]. Also for holes in Si a scalar value is used for the conductivity effective mass, it amounts to $m_{p,c} = 0.4\ m_0$ at 300 K. Both masses depend only very weakly on temperature.

Regarding the quasi-particle model for holes, the question is whether it implicates essentially different results compared with the previous classical picture of a void or bubble in the sea of bonding electrons. The answer is yes. Most evidently this is shown by the Hall-effect, measurements of which are the main experimental tool to investigate the basic processes underlying conductivity. In these experiments, a magnetic field $\vec{B}$ is applied perpendicular to a semiconductor strip in which a longitudinal current is flowing, the lateral voltage $V_H$ at the strip is measured, see Fig. 2.5.

Since the carriers are forced to flow along the strip, the Lorentz force $Q\ \vec{v} \times \vec{B}$ is balanced by the formation of a lateral electric field $\vec{\mathsf{E}}_H = -\vec{v} \times \vec{B}$ where $\vec{v}$ is the vector of (drift) velocity of the carriers and Q their charge. If x and z are the coordinates in the directions of $\vec{v}$ and $\vec{B}$, respectively, then the "Hall field" $\vec{\mathsf{E}}_H$ has the direction of the positive y coordinate. With the scalar components in this coordinate system denoted by normal letters (so that $\vec{v}$ = (v, 0, 0), $\vec{B}$ = (0, 0, B), $\vec{\mathsf{E}}_H$ = (0, $\mathsf{E}_H$, 0)), one obtains

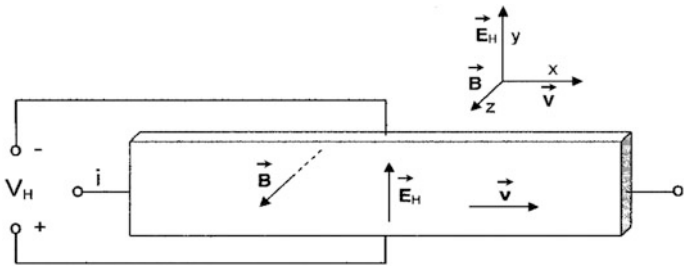



**Fig. 2.5** Basic arrangement for Hall measurements

$$E_H = v \cdot B = \frac{j}{Q \cdot C} B = R_H \cdot j \cdot B \quad \text{with} \quad R_H = \frac{1}{Q \cdot C} \tag{2.16}$$

Here the velocity was substituted by the current density according to $j = Q \cdot C \cdot v$ where $C$ denotes the carrier concentration. $R_H$ is called "Hall constant". If in a p-type semiconductor the hole current is in principle achieved by a motion of valence electrons as in the classical bubble model, the Hall constant would have the same negative sign as for an n-type specimen since then also $Q = -q$. Furthermore, because the concentration of valence electrons, $C$, is much higher than that of the empty states and also much higher than the concentration of electrons in an n-type semiconductor of equal conductivity, the absolute value of $R_H$ should be very small after the classical bubble model. Actually, however, the Hall constant measured for specimens doped with acceptors is positive in contrast to that of n-type samples and in magnitude it is comparable for both types of doping. Hence the holes manifest themselves as independent entities which in a magnetic field experience a Lorentz force like positively charged mass points. Hence p-type conductivity is a conductivity type of its own, equivalent to n-type conduction. Before the quantum theory of solids was developed, the positive Hall constant found in many specimens was a very irritating phenomenon.

## 2.5 The Doped Semiconductor

If in a silicon crystal some atoms in the lattice are replaced by atoms of an element of group V in the periodic table, e.g. phosphorus, each impurity atom has one electron more in the outer shell than necessary for the four covalent bonds. Therefore, one electron is only loosely bonded and needs only a small amount of energy—available probably as thermal energy—to be removed from the impurity atom and freed for conduction. These elements donating their fifth electron to the conduction band are called donors. The result is an extrinsic, n-type semiconductor with "n" pointing to the negative charge of the carriers. On the other hand, if the silicon atoms are replaced at some lattice points by atoms of an element of group III, e.g. boron, each impurity atom has one electron less than necessary for the four covalent bonds. Since the bonds between the impurity and the neighboring silicon atoms are nearly as tight as between the silicon atoms themselves, there is only little energy necessary for moving an electron out of a Si–Si bond in the neighborhood to the impurity to complete its bonds with the four silicon neighbors. Accepting an electron of the valence band and thus generating a mobile hole, these impurities leading to p-type conductivity are called acceptors. In the energy band picture, the donors have energy levels close to the conduction band in the bandgap, the acceptors close to the valence band edge. The energy levels of important dopants in Si and 4H-SiC are shown in Fig. 2.6.

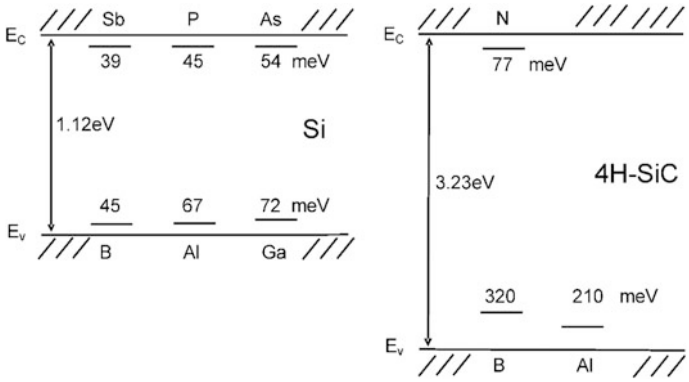



**Fig. 2.6** Energy levels of doping impurities in Si and 4H-SiC. The differences from the respective band are given, i.e. the ionization or activation energies for elevating an electron or hole from the neutral impurity atom into the conduction or valence band, respectively

In a quantitative model, impurities of group 3 and 5 are considered as hydrogen-like systems, the ground state energy of which represents the ionization energies $\Delta E_D = E_C - E_D$, $\Delta E_A = E_A - E_V$ [Smi59, Koh57]: In a neutral donor the electron not used for the lattice bonds is bound to the ion $D^+$ by the electrical field which is idealized to the Coulombic form $\mathsf{E} = q/4\pi\varepsilon r^2$ as in a homogeneous medium. Using for it the macroscopic dielectric constant $\varepsilon = \varepsilon_r \cdot \varepsilon_0$ of the semiconductor and for the electron its effective mass $m_n$ in the host crystal, the formulas of the hydrogen atom result, modified only by the numeric values of ε and $m_n$. Since the orbit radii obtained in this way are significantly larger than the distance between the semiconductor atoms, the assumptions are justified as a rough approximation. As ionization energy one obtains

$$\Delta E = \frac{m_n q^4}{8\varepsilon^2 h^2}$$

where $h$ is Planck's constant. For donors in Si ($\varepsilon = 11.7 \cdot \varepsilon_0$, $m_{eff} = 0.27 \cdot m_0$, $m_0$ free-electron mass) this yields: $\Delta E_D = 26.8\text{meV}$. Similarly, a neutral acceptor is understood as the ion $A^-$ (bound in the lattice by complete covalent bonds) surrounded by a hole in the idealized Coulomb field. With the hole effective mass of $0.4\ m_0$, the ionization energy of acceptors in silicon is obtained as $\Delta E_A = 40.0\,\text{meV}$. As is seen from Fig. 2.6, the experimental ionization energies are of the order of these model values, but somewhat higher. In SiC, the differences are larger. Apart from neglected effects of the immediate neighborhood of the impurity, the electron or hole assigned in the model to the orbit of the ion may be involved to some extent in the bonding of the ion in the lattice. Deep levels in the bandgap as shown by recombination centers (see Sect. 2.7.2) are beyond the range of this model.

Because of the small ionization energies and the many states in the conduction and valence bands available for electrons and holes, most donors and acceptors in Si

are ionized at room temperature. This follows from (2.1) or (2.2) if the Fermi level lies below the level of the donor or above the acceptor level in either case of doping. Considering the degree of ionization in detail we use the notation in the case of donor doping denoting the total impurity concentration with $N_D$ and the concentrations of neutral and ionized impurities with $N_D^0$, $N_D^+$, respectively. Then the number of occupied donor levels in (2.2) is $n_E = N_D^0$ and the number of unoccupied levels $N_E - n_E = N_D - N_D^0 = N_D^+$, hence the ratio of both, the occupation degree, is

$$\frac{N_D^0}{N_D^+} = g \cdot \mathrm{e}^{-\frac{E_D - E_F}{kT}} \tag{2.17}$$

where we have added a "degeneracy factor" $g$. This factor, which for donors is $g = 2$, is necessary because the neutral donors $D^0$ exist in two states depending on the spin orientation of the trapped electron. Nevertheless only one electron can be bound because for a second the Coulomb field is no longer present. Since the state $D^-$ does not appear, the Fermi energy does not include the degeneracy of the $D^0$ state and hence one has to take into account it separately [Spe58, Sho59]. The Fermi energy can be eliminated dividing Eq. (2.17) by (2.4), which yields:

$$\frac{N_D^0/N_D^+}{n/N_C} = g \cdot e^{\frac{E_c - E_D}{kT}} \tag{2.18}$$

As long as the intrinsic concentration and hence the hole density $p = n_i^2/n$ is small, the neutrality condition is $N_D^+ = n$ so that $N_D^0 = N_D - n$. Inserting this and solving for $n$, one obtains after simple conversion

$$n = \frac{N_D}{\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{N_D}{N_C} \cdot g \cdot e^{\frac{\Delta E_D}{kT}}}} \tag{2.19}$$

For small $\Delta E_D$ and $N_D << N_C$, this yields $n \approx N_D$ as expected. For acceptor doping the hole concentration is obtained exchanging $N_D$, $N_C$ and $\Delta E_D$ by the respective acceptor quantities $N_A$, $N_V$ and $\Delta E_A$ (see the more general Eq. (2.23) at $N_D = 0$). However, the degeneracy factor of acceptor levels in Si, Ge and SiC is $g = 4$ since there are two degenerate valence bands at $k = 0$, from which the levels are split off, and this results in a further degeneracy in addition to spin degeneracy [Bla62] [1]. For both n- and p-type conductivity, the degeneracy factor reduces the ionization ratio $n/N_D$ or $p/N_A$, respectively.

[1] Actually the occupation of excited states causes an additional degeneracy and enhancement of g [Bla62, p. 140 ff]. In Si this effect seems to be relative small owing to the high energy difference between the excited states and the ground state. Since a calculation under real conditions is not available, the effect is not taken into account.

As is confirmed by inserting the ionization energies of Fig. 2.6 and the effective density of states from (2.8), dopants in Si are nearly completely ionized at room temperature up to doping concentrations of $10^{17}$ cm$^{-3}$. Considerable de-ionization is obtained at higher concentrations. In this doping range one has to take into account, however, that the ionization energy decreases with increasing concentration starting from the values given in Fig. 2.6 for small densities. For a Ga impurity concentration of $5 \times 10^{17}$ cm$^{-3}$, which is about the maximum doping of the p-base in a thyristor, an activation energy of 65 meV was determined [Wfs60]. With this $\Delta E_A$ an ionization ratio of 66% of the total Ga concentration is obtained at 300 K. As function of temperature, the carrier concentration is shown in Fig. 2.7 for this example and additionally for phosphorus with a concentration of $10^{14}$ cm$^{-3}$. Whereas for the concentration $5 \times 10^{17}$ cm$^{-3}$ the deionization is noticeable up to 400 K, for a doping concentration of $10^{14}$ cm$^{-3}$ the ionization remains complete down to a temperature of 80 K. Above $T = 430$ K the intrinsic concentration becomes comparable with $N_D = 10^{14}$ cm$^{-3}$ and causes an increase of the carrier concentration.

To include this effect, the neutrality condition inserted in (2.18) has to take into account the minority carrier concentration. Writing $n = N_D^+ + n_i^2/n$ where in this range $N_D^+ = N_D$, one obtains

$$n = \frac{N_D}{2} + \sqrt{\left(\frac{N_D}{2}\right)^2 + n_i(T)^2} \tag{2.20}$$

In SiC the energy levels of dopants lie deeper in the bandgap (see Fig. 2.6), and hence a larger part of the doping atoms remains neutral, particularly for acceptors. For Al, the preferred acceptor dopant in SiC devices, the acceptor version of Eq. (2.19) with $N_V = 2.5 \times 10^{19}$ cm$^{-3}$ [Gol01] yields for a doping concentration $N_A = 1 \times 10^{16}$ cm$^{-3}$ an ionization ratio of only 35% at 300 K. This can strongly influence the device characteristics. Boron in SiC shows even a smaller degree of

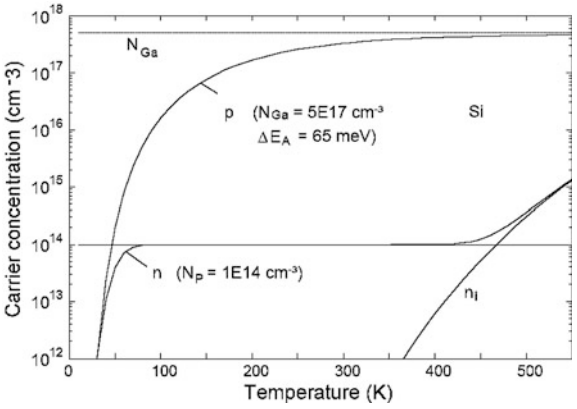



**Fig. 2.7** Carrier concentration as a function of temperature for a Ga doping of $5 \times 10^{17}$ cm$^{-3}$ and a phosphorus doping of $10^{14}$ cm$^{-3}$

ionization. Incomplete ionization is an obstacle for fabrication of layers with good emitter efficiency with SiC.

In GaN, silicon doping is the typical choice for intended n-doping. The activation energy of silicon in GaN is 5–9 meV, this allows effective doping [Qua08]. Mg is used as acceptor, its ionization energy is in the range 140–210 meV [Lev01], in [Qua08] a value of 173 meV is reported. Also native defects act as dopants in GaN, a vacancy on a nitrogen place $V_N$ acts as donor, a vacancy on a Ga-site $V_{Ga}$ is a shallow acceptor.

In diamond, the energy levels are even deeper in the bandgap [Gil79]. Boron is a deep acceptor level with activation energy of 370 meV [Gil79]. Phosphorus exhibits donor states with activation energies 0.8–1.16 eV [Oka90]. Therefore, the ionization ratio at room temperature will be low, and it is difficult to reach a significant carrier density for current transport.

Often it is important to know the Fermi level for a given doping density. For complete ionization and temperatures where $n_i << N_D$, $N_A$, the Fermi-level is related in a simple manner to the doping concentration. If we restrict again to the nondegenerate case, one obtains inserting $n = N_D$ in (2.4):

$$E_C - E_F = k \cdot T \cdot \ln\left(\frac{N_C}{N_D}\right) \tag{2.21}$$

For silicon this and the analogous equation for acceptor doping are plotted in Fig. 2.8 for 300 and 400 K. The Fermi energy is a linear function of the logarithm of the doping density.

According to Eq. (2.19) the ratio $n/N_D$ decreases with increasing doping density $N_D$ and, assuming $\Delta E_D$ as constant, becomes small at high $N_D$. Really this is not observed, on the contrary at doping concentrations of $10^{19}$ cm$^{-3}$ and higher the impurities are completely ionized. This is explained by the mentioned decrease of the ionization energy and other high doping effects. The decrease of $\Delta E_D$, $\Delta E_A$ is caused by screening of the impurity charge by free carriers, e.g. $D^+$ by electrons, as well as by the formation of a tail of states at the neighbored band, because the

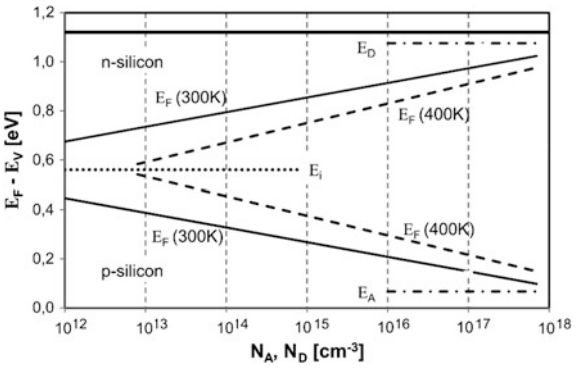



**Fig. 2.8** Fermi level in n- and p-type Si versus doping concentration at $T = 300$ K

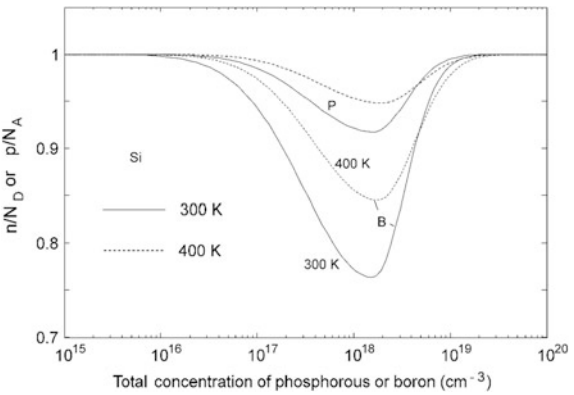



**Fig. 2.9** Ionization ratio of phosphorus and boron in Si at 300 and 400 K as a function of the total concentration of the respective impurity. After Kuzmicz [Kuz86]

periodic lattice potential is disturbed by the Coulomb field of the statistically distributed impurities. Additionally, the levels of the impurity atoms spread and form an impurity band because the wave functions of the bound impurity states overlap at high concentrations and hence the levels split up. These effects begin more than an order of magnitude below the concentration $N_{C,V}/2$, where degeneracy sets in. Since they are not yet completely mastered theoretically from first principles, practical approaches use semi-theoretical or empirical modelling concepts with adjusted parameters. A calculation of Kuzmicz [Kuz86], who considers the mentioned impurity band and the decrease of the mean ionization energy, yielded the dependence of the ionization ratio on impurity concentration as shown in Fig. 2.9.

The figure refers to phosphorus and boron in Si, the most commonly used dopants. The curves were recalculated using analytical expressions developed in [Kuz86] on the base of numerical calculations. The deionization has its maximum near $2 \times 10^{18}$ cm$^{-3}$ and amounts to 9% for phosphorus and 23% for boron at 300 K. Above $10^{19}$ cm$^{-3}$, the ionization is again complete. The deionization in the intermediate range becomes noticeable in the temperature dependence of the resistivity.

What if both donors and acceptors are present simultaneously? This is always the case in devices doped by impurity diffusion and particularly in the vicinity of diffused pn-junctions, which are defined by $N_A = N_D$. To see the influence of this compensation we consider the case that the acceptor doping predominates and the difference $N_A - N_D$ is large against the intrinsic concentration $n_i$. Due to the deep lying Fermi level, all the donors are then ionized, $N_D^+ = N_D$. The hole concentration in a neutral region is hence given by $p = N_A^- - N_D$. Usually it is assumed that also the acceptors are completely ionized, so the hole density is equal to the net impurity concentration, $p = N_A - N_D$. Taking the deionization of the acceptors into account, one obtains for the ionization ratio of the *net* acceptor concentration $N_{A,net} = N_A - N_D$ with $\mathrm{r} \equiv N_A^-/N_A$:

$$\frac{p}{N_{A,net}} = \frac{rN_A - N_D}{N_{A,net}} = r - (1-r)\frac{N_D}{N_{A,net}}$$

Related to the reduced net acceptor doping the deionization of the acceptors becomes a higher weight, hence $p/N_{A,net}$ decreases with compensating doping $N_D$. But also the acceptor ionization ratio r itself depends on the concentration $N_D$. Expressing $N_A^-/N_A^0$ by Eq. (2.2) and multiplying with (2.5) one obtains

$$\frac{N_A^-}{N_A^0}p = \frac{N_V}{g}\exp\left(-\frac{E_A - E_V}{kT}\right) \equiv C \tag{2.22}$$

Again a degeneracy factor $g$ is included to allow for the enhanced occupation probability of the neutral impurity state. The concentration $C$ is the mass law constant of the reaction $N_A^- + p \Leftrightarrow N_A^0$. Since now p is reduced by the compensating dopant, the occupation degree $\gamma \equiv N_A^-/N_A^0$ is enhanced and thus also the ionization ratio $r = \gamma/(1+\gamma)$. Inserting the neutrality condition $N_A^- = p + N_D$ and the equation $N_A^0 = N_A - N_A^- = N_A - p - N_D$ into (2.22) one obtains for the hole density

$$\frac{(p+N_D)p}{N_{A,net} - p} = C \tag{2.23}$$

This has the solution

$$p = \frac{2N_{A,net}}{1 + \frac{N_D}{C} + \sqrt{\left(1 + \frac{N_D}{C}\right)^2 + 4\frac{N_{A,net}}{C}}} \tag{2.24}$$

Equation (2.24) is a generalization of the acceptor version of (2.19), into which it turns with $N_D = 0$. The equation shows explicitly how the fraction of the ionized net acceptor doping depends, apart from $N_{A,net} = N_A - N_D$ itself, on the compensating dopant concentration $N_D$. Because $C$ is of the order the magnitude $10^{18}$ to $10^{19}$ cm$^{-3}$, the effect is only significant for $N_D \gtrsim 5 \times 10^{16}$cm$^{-3}$, hence for a realistic description one has to consider high-doping effects. For not too high concentrations these can be expressed approximately by an effective decrease of the ionization energy $\Delta E_A = E_A - E_V$ with doping densities. Since this leads to an increase of the concentration C according to (2.22), the decrease of p with $N_D$ as given by (2.24) is strongly weakened. The ionization energy of boron was reported to depend on doping densities as $\Delta E_A = 46 - 3 \times 10^{-5} \cdot (N_A^+ + N_D)^{1/3}$ meV [Li78]. In Fig. 2.10 the dependence of $p = N_{A,net}^-$ on $N_D$ obtained with this $\Delta E_A$ is plotted for two temperatures. The net acceptor density $N_A - N_D$ is taken to be constant and equal to $1 \times 10^{16}$ cm$^{-3}$. At 300 K the ionization ratio decreases to 80% up to $N_D = 2 \times 10^{18}$ cm$^{-3}$. Hence the deionization in regions of strong compensation is considerable already at low net doping densities. For absent compensation, $N_D = 0$, the acceptor ionization ratio obtained from (2.24) with the stated $\Delta E_A(N_A)$ is up to

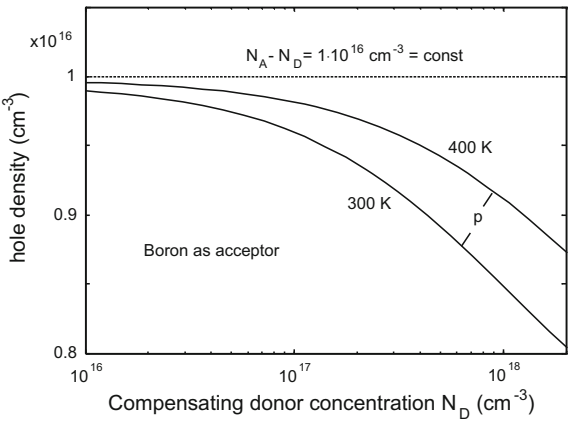



**Fig. 2.10** Hole concentration for a constant net acceptor doping $N_A - N_D$ versus compensating donor density $N_D$

nearly $N_A = 2 \times 10^{18}$ cm$^{-3}$ in rough accordance with the curves for boron in Fig. 2.9, beyond this doping the approach fails. In literature [Bla62, p. 132 ff] doping with more than one impurity species is treated extensively using equations like (2.24).

Till now we have assumed that the semiconductor is neutral and in thermal equilibrium. In devices, however, there are regions with space charge where carriers are depleted and, on the other hand, regions with injected carriers adding to those supplied by dopants. Departing from above calculations, the ionization in space charge regions is nearly always complete because the depletion of carriers ($p << N_A^-$) shifts the reaction $A \Leftrightarrow A^- + \oplus$ to higher $N_A^-$. Under conditions of high injection, however, the deionization is enhanced, and especially in this case it can affect device operation. These effects can be calculated quantitatively from Eqs. (2.18) and (2.22).

In spite of the limits shown by this discussion, the doping atoms in silicon are usually to a fairly high degree ionized at operating temperatures $T \gtrsim 250\,K$. In fact *most analytical calculations and programs for device simulation are just based on the assumption of complete ionization throughout the device and during the whole switching time considered.* The above discussion shows, where a more exact treatment may be worth the enhanced effort. For devices in SiC and GaN this is often the case.

The high doping effects mentioned are responsible for another phenomenon which influences the characteristics of power devices: the reduction of bandgap and the associated increase of the intrinsic concentration $n_i$ at high doping densities. From measurements on the p base of bipolar transistors Slotboom and De Graaff [Slo76] obtained the following empirical relationship for the bandgap narrowing, $\Delta E_g$, in dependence of doping concentration $N$:

$$\begin{aligned} \Delta E_g &= 9 \times 10^{-3}\text{eV} \cdot \left( \ln \frac{N}{10^{17}\text{cm}^{-3}} + \sqrt{(\ln \frac{N}{10^{17}\text{cm}^{-3}})^2 + 0.5} \right) \\ &\approx 18\text{meV} \cdot \ln \frac{N}{10^{17}\text{cm}^{-3}} \quad \text{for N} > 5 \times 10^{17}\text{cm}^{-3} \end{aligned} \tag{2.25}$$

Calculations considering band tails and impurity band were found to be in rough accordance with (2.25) [Slo77]. The dependency observed for regions with acceptor doping is used also for highly doped n regions. Another effect which has been shown by Lanyon and Tuft [Lan79] to result in an effective reduction of bandgap is the electrostatic field energy required when an electron-hole pair is created within the surrounding of majority carriers. This stored energy included in the band gap becomes smaller with increasing (majority) carrier density due to screening. Based on this mechanism, the following theoretical formula has been derived in the range where Maxwell-Boltzmann statistics apply [Lan79]

$$\Delta E_g = \frac{3q^2}{16\pi\varepsilon} \left( \frac{q^2 n}{\varepsilon kT} \right)^{1/2} = 22.6 \left( \frac{n}{10^{18}} \frac{300}{T} \right)^{1/2} \text{meV} \tag{2.26}$$

$\Delta E_g$ depends here on the carrier concentration n (in case of n doping) rather than the doping concentration, which can be of significance especially in space charge regions. The numerical expression on the right hand side of (2.26) is obtained for Si using the dielectric constant $\varepsilon = 11.7 \cdot \varepsilon_0$. Although band tails and impurity band are not considered, (2.26) is also claimed to describe largely the measurements which were partly reinterpreted [Lan79]. Differing from the results of Slotboom and De Graaff, (2.25), the bandgap reduction according to (2.26) depends on temperature.

A plot of the two relationships is shown in Fig. 2.11. Around $10^{18}$ cm$^{-3}$, the equation of Lanyon and Tuft yields a considerably smaller bandgap narrowing than that of Slotboom and De Graaff, above $2 \times 10^{19}$ cm$^{-3}$ it is the other way round. Both evaluations are not free of assumptions; the experimental result (2.25) is supported theoretically in [Slo77].

The primarily determined quantity in [Slo76] is the intrinsic carrier concentration $n_i$ in dependence on doping concentration. For conversion to bandgap narrowing the effective density of states $N_V$ or $N_C$ of the relevant band is assumed to be unchanged. Hence as obtained by inserting the reduced bandgap $E_g = E_{g0} - \Delta E_g$ into (2.6), the following relation is used for conversion between $n_i$ and $\Delta E_g$:

$$n \cdot p = n_i^2 = N_C N_V e^{\frac{-Eg(N)}{kT}} = n_{i0}^2 e^{\frac{\Delta Eg}{kT}}, \tag{2.27}$$

where $n_{i0}$ denotes the intrinsic concentration at low doping (or carrier) concentrations corresponding to the bandgap $E_{g0}$. We will use these results in later Sects. 3.4 and 5.4.5 for the injection efficiency, i.e. the influence of recombination in highly doped emitter regions on device characteristics.

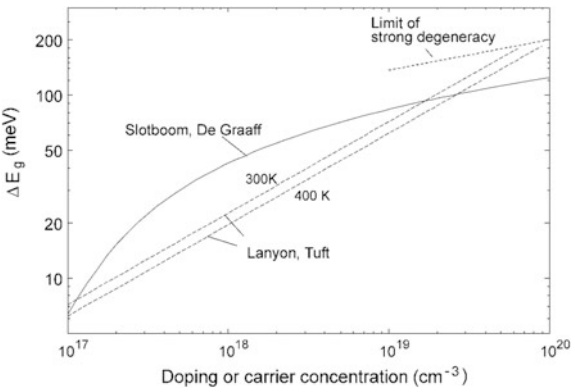



**Fig. 2.11** Bandgap narrowing as a function of doping or carrier density according to [Slo76], Eq. (2.25) (solid line), and [Lan79], Eq. (2.26) (dashed lines), respectively. The dotted lines represents a limiting expression for strong degeneracy [Lan79]

At the end of this paragraph we mention some frequently used notations. The terms *majority carriers* and *minority carriers* are used to identify the carriers of the type supplied by the (dominating) doping atoms and the carriers of the second kind, which are in minority and related with the first by the mass law equation $n \cdot p = n_i^2$ in thermal equilibrium. In this case, the density of minority carriers is in silicon many orders of magnitude smaller than the majority concentration. For a donor density $N_D = 1 \times 10^{14}$ cm$^{-3}$ one obtains at room temperature with $n_i \approx 10^{10}$ cm$^{-3}$ a majority carrier concentration $n = 1 \times 10^{14}$ cm$^{-3}$ and a minority concentration $p = n_i^2/n = 1 \times 10^6$ cm$^{-3}$. At higher temperatures, the minority carrier concentration can be calculated inserting (2.20) into $p = n_i^2/n$, see also Fig. 2.7. Whereas in thermal equilibrium the minority carriers are insignificant for the properties of bulk material, they play an essential part for the functioning of pn-devices based on injection and extraction of minority carriers. This will be investigated later.

In devices with layers of different doping levels, a superscript index '−' or '+' is added to the symbols for the conductivity type to indicate the doping level. So $n^-$, $p^-$ signify n and p layers with a much lower doping than neighboring regions; likewise $n^+$, $p^+$ mean layers with a much higher doping level. Typically the doping ranges indicated in this manner are as follows:

| | |
|---|---|
| $n^-, p^-$ | $10^{12}-10^{14}$cm$^{-3}$ |
| n, p | $10^{15}-10^{18}$cm$^{-3}$ |
| $n^+, p^+$ | $10^{19}-10^{21}$cm$^{-3}$ |

Normally a power device contains in the interior a weakly doped $n^-$ region which essentially determines the characteristics. Heavily doped $n^+$ and $p^+$ layers border on the metalized surfaces.

## 2.6 Current Transport

### 2.6.1 Carrier Mobilities and Field Currents

As discussed in Sects. 2.3 and 2.4, electrons and holes in the semiconductor behave essentially like free particles which, however, are scattered by vibrating lattice atoms, impurity ions and other scattering centers. Like molecules in a gas they have a kinetic thermal energy which on statistical average is

$$E_{kin} = \frac{m}{2} v_{th}^2 = \frac{3}{2} kT \tag{2.28}$$

where $v_{th}$ is the mean thermal velocity and m the effective mass of the respective carrier. Already at room temperature the mean thermal velocity is very high: With the electron effective mass $m = m_n = 0.27 \cdot m_0$, one obtains from (2.28) $v_{th} = \sqrt{(3kT/m_n)} = 2.2 \times 10^7$ cm/s = 220 µm/ns at 300 K. The mean free path between two scattering events is of the order 10 nm in lightly doped Si and still smaller if many impurities are present. Since the thermal motion is statistically distributed over all directions, the current resulting from the large number of carriers is zero between terminals on equal potential. The thermal velocity used in (2.28) is defined as the root of the mean value of the squared velocity, a rough measure for the mean *absolute* velocity. Contrary to it, the normal *linear* mean value of the thermal velocities taking account also of their directions is zero for both electrons and holes, if no field is present.

When an electric field $\mathsf{E}$ is applied, each carrier experiences a force $\pm\, q \cdot \mathsf{E}$ and is accelerated between two collisions. Hence the thermal velocity is superimposed by an additional velocity which for holes has the direction of the electric field and for electrons the opposite. Averaging linearly over time and the carriers of each type, now *non-zero* mean velocities $v_n$, $v_p$ of the electrons and holes result. These velocities caused by the field are called *drift* velocities. For low fields, defined by the condition that the drift velocities are small compared with the thermal velocity $v_{th}$, the mean free time $\tau_c$ between two collisions which depend on the total velocity is independent of the field. Therefore, in this range, the drift velocities are proportional to the field strength

$$v_{n,p} = \mp \mu_{n,p} \cdot \mathsf{E} \tag{2.29}$$

The proportionality factors $\mu_n$ for electrons and $\mu_p$ for holes are called *mobilities*. Due to the minus sign used in the case of electrons both mobilities are positive constants. Inserting (2.29) into the condition $v << v_{th}$, and omitting now the indices n or p, one obtains as condition which the field must satisfy for constant mobilities

$$\mathsf{E} << \frac{v_{th}}{\mu} \tag{2.30}$$

The significance of the mobilities follows from their connection with the macroscopic electric current densities. These are obtained from (2.29) as

$$\begin{aligned} j_n &= -q \cdot n \cdot v_n = q \cdot \mu_n \cdot n \cdot \mathsf{E} \\ j_p &= q \cdot p \cdot v_p = q \cdot \mu_p \cdot p \cdot \mathsf{E} \end{aligned} \tag{2.31}$$

with the concentrations $n$ and $p$ of electrons and holes. The sum of both is the total current density $j$:

$$\begin{aligned} j &= j_n + j_p = q \cdot (\mu_n \cdot n + \mu_p \cdot p) \cdot \mathsf{E} \\ &= \sigma \cdot \mathsf{E} = \mathsf{E}/\rho \end{aligned} \tag{2.32}$$

where

$$\sigma \equiv q \cdot (n \cdot \mu_n + p \cdot \mu_p) = 1/\rho \tag{2.33}$$

is the electrical conductivity and $\rho$ the resistivity. According to these equations, the mobilities are material parameters which determine the ohmic voltage drop $V = \mathsf{E} \cdot \Delta x = \rho \cdot j \cdot \Delta x$ for a given current density and hence the power loss density $V \cdot j$ and heat generation. Hence, the mobilities determine the maximum allowed current density of devices. Together with other characteristics they decide on the suitability of a semiconductor for power devices.

An overview of mobilities in semiconductors is given in Table 2.2. It shows at first that the mobility of holes is in all cases significantly smaller than that of electrons. Hence, n and p regions in devices are not equivalent. Especially in unipolar devices which conduct current only by majority carriers, the weakly doped region required for the blocking voltage is chosen preferentially of n type doping. As mentioned in Sect. 2.1, the very high electron mobility of GaAs offers the possibility to make Schottky diodes with a low on-state voltage drop and simultaneously a relative large thickness necessary for a high blocking voltage. For the

**Table 2.2** Mobilities of various semiconductors at room temperature for light doping. For 4H-SiC the mobilities parallel to the hexagonal axis are given, for GaN (2H-type) the mobilities perpendicular to the hexagonal axis in bulk material and additionally the electron mobility in the mentioned 2-dimensional electron gas (2DEG). The table contains also the saturation drift velocity of electrons, a property discussed at the end of this chapter

| | $\mu_n$ [cm$^2$/(Vs)] | $\mu_p$ [cm$^2$/(Vs)] | $v_{sat(n)}$ [cm/s] |
|---|---|---|---|
| Ge | 3900 | 1900 | $6 \times 10^6$ |
| Si | 1420 | 470 | $1.05 \times 10^7$ |
| GaAs | 8000 | 400 | $1 \times 10^7$ |
| 4H-SiC | 1000 | 115 | $2 \times 10^7$ |
| GaN | 990 | 150 | $2.5 \times 10^7$ |
| GaN 2DEG | up to 2000 | | |
| Diamond | 2200 | 1800 | $2.7 \times 10^7$ |

hexagonal semiconductors SiC and GaN the mobilities are anisotropic, i.e. they are different in directions parallel ($\mu_{||}$) and perpendicular ($\mu_{\perp}$) to the hexagonal axis. The values in the table are the mobilities parallel to the hexagonal axis, which in vertical devices is the direction of main current flow. In 4H-Si the anisotropy is small, $\mu_{n\|} \approx 1.2\ \mu_{n\perp}$ [Scr94] in contrast to the polytype 6H-SiC which electron mobility $\mu_{n||}$ is about a factor 5 smaller than $\mu_{n\perp}$ and also than $\mu_{n\|}$ in 4H-SiC. This is the main reason why 4H-SiC is preferred now against 6H-SiC, which polytype attracted much research efforts in the 1990s. To judge different semiconductors regarding conduction losses in devices, one has to use the mobilities together with other relevant properties such as the necessary thickness and doping density allowed for a given blocking voltage. Although the mobilities of 4H-SiC are lower than those of Si, in combination with the much smaller thickness and higher allowed doping of the base region a much smaller on-state resistance of unipolar devices can be achieved. This holds also in comparison with GaAs. GaN is with respect to the mobilities comparable to SiC, however, in a two-dimensional electron gas (2DEG) at a heterojunction to AlGaN electron mobilites up to 2000 $cm^2V^{-1}s^{-1}$ are measured, see Chap. 4. High mobilities are measured in diamond, but because of large ionization energies of the dopants the carrier concentration is small.

The mobilities are constant with regard to the field (if small enough), but depend on doping concentration and temperature. An accurate knowledge of their dependence on doping concentration is very important particularly because the mobility determines the relation between resistivity and doping density which is used daily to conclude from the simply measurable resistivity to doping density N. Assuming complete ionization one has for an n type wafer

$$\rho(N) = \frac{1}{q \cdot \mu_n \cdot n} = \frac{1}{q \cdot \mu_n(N)} \cdot \frac{1}{N} \tag{2.34}$$

In an undoped and weakly doped semiconductor, the mobilities are determined by scattering on phonons, i.e. vibrating lattice atoms. Above a doping concentration of $10^{15}$ $cm^{-3}$, the mobilities are noticeably and at higher concentrations strongly reduced by collisions with doping ions. At still higher concentrations the scattering by ions is limited by the carriers themselves by screening of the impurity charge. The experimental dependence of mobilities in Si on the donor or acceptor ion concentration, respectively (= majority carrier concentration) is depicted in Fig. 2.12 which also shows fitting curves to the measurements. The figure is valid at room temperature. The experimental curves represent measurements of Thurber and coworkers on phosphorus doped silicon for $\mu_n$ [Thu80a] and boron doped silicon for $\mu_p$ [Thu80b] using their analytical representation of measurements. For $n > 8 \times 10^{18}$ $cm^{-3}$ an empirical dependence for the electron mobility of Masetti et al. [Mas83] is shown, who determined the mobilities at very high doping levels. The dashed lines are fits to the experiments by the often used formula of Caughey and Thomas [Cau67]:

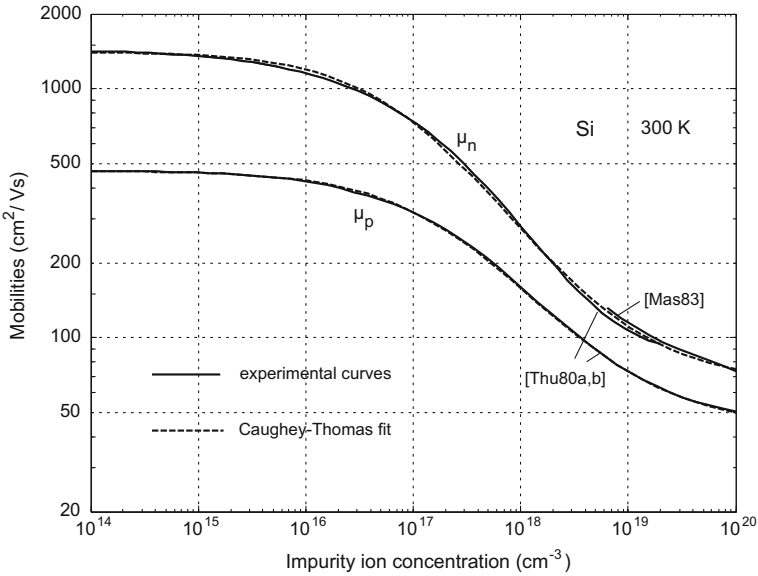



**Fig. 2.12** Mobilities in Si as function of impurity ion density

$$\mu = \mu_\infty + \frac{\mu_0 - \mu_\infty}{1 + \left(N/N_{ref}\right)^\gamma} \tag{2.35}$$

The limiting values $\mu_0$ and $\mu_\infty$ for low and high concentrations and the concentration $N_{ref}$ at which the mobility adopts the mean value between them, are matched to the experiments. $N$ denotes the impurity ion concentration. Parameter values used for the fit are given in Appendix A, where also their variation with temperature is given. As is seen, this simple approach is well suited to describe the experimental dependence up to impurity densities of $1 \times 10^{20}$ cm$^{-3}$. Around $2 \times 10^{18}$ cm$^{-3}$ the carrier density deviates noticeably from the total impurity density as discussed in the previous paragraph. To obtain the mobility for a given total impurity concentration in this range, first the ionized impurity density has to be determined using Fig. 2.8. Although Eq. (2.35) can be used also with N defined as total impurity density, the plot versus ion density is preferred, because the ionized impurity concentration is the quantity determining the mobilities, while scattering by impurities in the neutral state may be neglected. Hence the plot is applicable with good accuracy also for doping with As, Ga and Al, which are ionized to a different extent. Often complete ionization is assumed when discussing mobilities in the range of operation temperatures of devices. We will also do this in what follows.

Also for silicon carbide the Caughey-Thomas formula (2.35) can be used to describe the dependency of the mobilities on the doping density. The parameters for 4H-SiC are given in Appendix A as well.

So far we have dealt with the mobilities of majority carriers in thermal equilibrium. During on-state of power devices high concentrations of electrons and holes are injected into the weakly doped base region. Hence electrons are scattered

by holes and holes by electrons (electron-hole scattering), and this leads to a similar decrease of the mobilities as scattering by ions. The voltage drop over the base region is strongly dependent on this effect. The mobility sum $\mu_n + \mu_p$ at high injection levels where the electron and hole concentrations are equal to each other ($n \approx p \gg N_{dop}$), has been measured as function of the carrier concentration by Dannhäuser and Krause [Dan72, Kra72]. Figure 2.13 shows their experimental dependency. Theoretically, electron-hole scattering is often described by a classical formula of Fletcher [Fle57] using for combination with the lattice scattering mobility an expression of Debye and Conwell [Deb54]. The mobility sum as function of n obtained from this theory at 300 K is shown in Fig. 2.13 also. Considering that the effective masses of electrons and holes entering the formulae were not matched, the agreement with the measurements is good. A more general mobility model using quantum mechanical scattering theory has been developed by Klaassen [Kla92]. The result of this model for $\mu_n + \mu_p$ is plotted in Fig. 2.13 additionally. The curve runs close to the former ones, except above $3 \times 10^{17}$ cm$^{-3}$ where the result of Klaassen becomes considerably higher. The difference against the measurements is explained in [Kla92] with heating of the samples by the current pulses used in the measurements of [Dan72, Kra72]. However, also simplifying assumptions of the theory may be a cause for the discrepancy.

Since the mobility theory is very complicated, a simplified description can be useful for a survey. For this purpose it is of interest how the mobilities at high injection levels as function of the carrier concentration compare really with the dependency on the doping density in thermal equilibrium. To show this, the quantity $\mu_n + \mu_p$ obtained from Eq. (2.35) with N replaced by the carrier concentration n = p is plotted in Fig. 2.13 additionally. Again the parameter sets for $\mu_n$ and

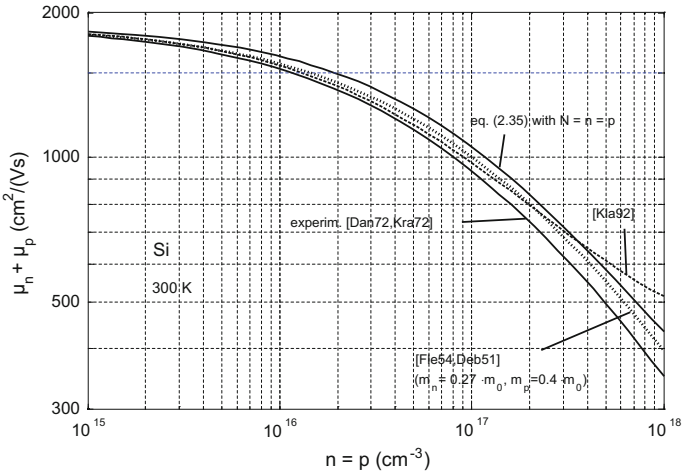



**Fig. 2.13** Mobility sum at high injection levels as a function of carrier concentration

$\mu_p$ as given in Appendix A are used. As is seen, the obtained mobility sum as function of the carrier concentration differs not much from the other curves. If the difference against the experimental dependency proves true in view of a possible heating and the spread of the measurements (about ± 10% around the fitting curve), electron-hole scattering is somewhat more effective than ion scattering.

In a rough approximation, however, the measurements suggest that impurity ions in thermal equilibrium and carriers at high injection levels are nearly equally effective in reducing the mobility sum. This will hold hence also for the single mobilities. Since scattering at the two types of scatterers, ions and carriers with opposite charge, may be assumed to be independent of one another, the (approximate) equivalence follows also for a mixture of both. Hence defining $N$ in (2.35) as the total concentration of scatterers, $N = N_D + p$ in the $\mu_n(N)$ expression and $N = N_A + n$ in $\mu_p(N)$, the equation describes the effect of impurity *and* electron-hole scattering. The electron mobility in an n-region and the hole mobility in a p region each with arbitrary minority carrier injection level is obtained hence as:

$$\mu_n = f_n(N_D + p) \tag{2.36a}$$

$$\mu_p = f_p(N_A + n) \tag{2.36b}$$

where $f_n(N)$ and $f_p(N)$ are the function (2.35) with the $\mu_n$ and $\mu_p$ parameter set, respectively. Scattering of electrons by electrons and of holes by holes has not been considered, since their influence is of second order.[2] Like carrier scattering is included nevertheless to a significant part, because it is contained in the used empirical relationship for thermal equilibrium and the reproduced dependency at high injection level. The range of approximate validity of (2.36a, b) reaches to minority carrier concentrations of about $5 \times 10^{17}$ cm$^{-3}$, as pointed out below.

The model can be extended to the presence of a compensating doping. The mobilities are influenced then in addition to attractive also by repulsive impurity ions. Although of considerable practical interest, experimental results on mobilities in compensated semiconductors are hardly available. Theoretically, repulsive Coulomb fields have (approximately) the same scattering cross section and hence impact on the mobilities as attractive [Mol64, Smi59, Kla92], provided the concentrations are smaller than about $5 \times 10^{17}$ cm$^{-3}$ at normal operating temperatures. This restriction occurs because the carrier velocities and the minimum distances of the scattered particles from the Coulomb centers must not be too small on average. Hence, with this limitation of the concentration of repulsive ions, the mobilities in compensated regions in thermal equilibrium are given by (2.35) with the

[2]Like-carrier scattering does not influence the mobilities directly, because the total momentum of two colliding particles and hence the current they transport are not changed by the impact. Nevertheless it reduces the mobilities by randomizing the velocity distribution of carriers, which makes scattering by ions and phonons more effective [Deb54, Luo71]. The influence of like-carrier scattering on the effectiveness of *electron-hole* scattering is unclear till now.

concentration N defined as the sum of donor and acceptor concentrations, $N = N_D + N_A$. Generally, if also injected carriers are present, the mobilities are given by

$$\mu_n = f_n(N_D + N_A + p) \tag{2.37a}$$

$$\mu_p = f_p(N_A + N_D + n) \tag{2.37b}$$

where $f_n(N)$ and $f_p(N)$ are again the function (2.35) with the $\mu_n$ respectively $\mu_p$ parameter set. The mobilities are determined by the sum of the doping concentrations (while the majority carrier concentration equals the positive *difference*). (2.37a, b) is applicable also to the mobility of minority carriers. For example the hole mobility in an n-region with $N_A = 0$ follows from (2.37a) in thermal equilibrium as $\mu_p = f_p(N_D + n) = f_p(2N_D)$. Because of the additional scattering by majority carriers the minority carrier mobility is smaller than the mobility of majority carriers for equal doping concentration.

At higher concentrations, however, the scattering cross section of repulsive ions is considerably smaller than for attractive Coulomb fields, as is obtained from a rigorous quantum mechanical scattering theory [Kla92]. Also electron-hole scattering becomes less effective at high carrier concentrations. These effects can be taken into account in (2.37a, b) supplying the concentrations of repulsive impurity ions and of scattering carriers with weighting factors which are smaller than 1. One consequence is that the minority carrier mobilities at doping densities higher than about $2 \times 10^{18}$ cm$^{-3}$ are found to be *higher* than the respective majority mobilities, although electron-hole scattering adds to repulsive impurity scattering. Below about $2 \times 10^{17}$ cm$^{-3}$ on the other hand, the weighting factor of the carrier concentrations becomes somewhat higher than 1, reaching about 1.4 [Kla92]. This can be a cause of the deviation of the dependency obtained from (2.35) from the experimental curve in Fig. 2.13. As mentioned, another high concentration effect is screening of the Coulomb centers by oppositely charged carriers. In contrast to the empirical fitting Eq. (2.35), the extensions (2.36a, b), (2.37a, b) do not include screening effects produced by injection and compensation at high concentrations. Injection enhances and compensation lowers screening and mobilities as compared with (2.37a, b). The model of Klaassen [Kla92] includes screening in a rough manner. Like carrier scattering is neglected in the theoretical part of this model.

The *temperature* dependence of mobilities is determined by the same effects as the dependence on doping density. At *low* doping, scattering on thermal lattice vibrations predominates and yields a decrease of mobilities with temperature approximately proportional to $T^{-2}$. Impurity scattering is shown theoretically to yield a mobility $\mu_i$ *increasing* with temperature nearly as $T^{3/2}$, if screening is neglected. This comes from the increase of the thermal velocity causing a decrease of scattering by the Coulomb centers. The increase of thermal velocity, however, implicates also a decrease of screening which at very high doping density inverts the temperature dependence of $\mu_i$ to a slow *decrease* with $T$. These effects of the

partial mobilities $\mu_l$ and $\mu_i$ find themselves also in the total mobility $\mu$ which is formed according to the approximate Matthiesen rule

$$\frac{1}{\mu} = \frac{1}{\mu_l} + \frac{1}{\mu_i} \tag{2.38}$$

Considering the temperature range 250 – 450 K, the result is that at low doping the (total) mobilities in Si show a strong decrease with increasing T, an approximate independency on T in the doping range $10^{18}$ cm$^{-3}$ to $10^{19}$ cm$^{-3}$ and at very high doping a minor decrease. The experimental temperature dependence can be well described – together with doping dependence – using temperature-dependent parameters $\mu_0$, $\mu\infty$, $N_{ref}$ and $\gamma$ in formula (2.35) as given in Appendix A.

In Fig. 2.14 the temperature dependency of $\mu_n$ and $\mu_p$ in silicon are shown for two doping densities, one which is typical for the n-base region of power devices (left hand side) and the other for the p-base in thyristors, IGBTs and bipolar transistors (right hand side). The temperature dependence at $3 \times 10^{17}$ cm$^{-3}$ is considerably weaker than at $1 \times 10^{14}$ cm$^{-3}$. That the resistivity $\rho = 1/(q\mu_p N_A)$ of the p base of mentioned devices increases relatively weakly with $T$, is very desirable for the *lateral* resistance of the $p$ base (see Chaps. 8 and 10). Particularly the on-state voltage drop including the on-resistance of MOSFETs depends essentially on the mobilities and their temperature dependence. Equations (2.36a, b), (2.37a, b) together with the parameter set of Appendix A provide also a model for the temperature dependence of the mobilities at high injections and in the case of compensation.

## 2.6.2 High-Field Drift Velocities

At high electric fields, where condition (2.30) is not satisfied, the drift velocity is no longer proportional to the field, but increases weaker. At very high field strength it

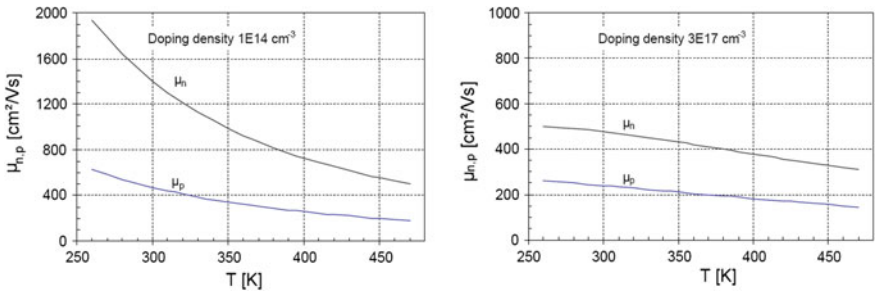



**Fig. 2.14** Temperature dependence of mobilities in Si at two doping densities (see text)

approaches a limiting value, the saturation drift velocity $v_{sat}$. The dependency of the drift velocities of electrons and holes on the electric field is often expressed in the form [Cau67, Tho80]:

$$v_{n,p} = \frac{\mu_{n,p}^{(0)} \cdot \mathsf{E}}{\left(1 + \left(\frac{\mu_{n,p}^{(0)} \cdot \mathsf{E}}{v_{sat(n,p)}}\right)^{\beta}\right)^{\frac{1}{\beta}}} \tag{2.39}$$

where $\mu_{n,p}^{(0)}$ are the low-field mobilities discussed above.[3] The additional marking is introduced, because Eq. (2.29) is used also for the non-linear range of fields where the mobilities $\mu_{n,p} \equiv v_{n,p}/\mathsf{E}$ are field-dependent. For small $\mathsf{E}$, (2.39) turns into (2.29) with $\mu_{n,p} = \mu_{n,p}^{(0)}$. The exponent $\beta$ was proposed by Caughy and Thomas [Cau67] to be $\beta_n = 2$ for electrons and $\beta_p = 1$ for holes. Jacobini and coworkers [Jac77] have determined the $\beta$'s renewed and as functions of temperature. They obtained $\beta_n = 2.57 \times 10^{-2} \times T^{0.66}$, $\beta_p = 0.46 \times T^{0.17}$, which yields a value near 1 at 300 K in *both* cases. Besides $\mu_{n,p}$, and $\beta_{n,p}$, also the saturation velocities are functions of temperature. According to Jacobini et al. [Jac77] they decrease with $T$ as $v_{sat(n)} = 1.53 \times 10^9/T^{0.87}$, $v_{sat(p)} = 1.62 \times 10^8/T^{0.52}$ in cm/s.

Using these results in (2.39), one obtains at 300 K the field dependences shown in Fig. 2.15. Below $10^3$ V/cm, fields which occur in devices during the conducting state, one obtains the linear dependency described by constant mobilities. Soon above this value, however, the increase of $v_n$ and somewhat later that of $v_p$ becomes sublinear, as is expected also from condition (2.30). At $3 \times 10^4$ V/cm the drift velocities approach the respective saturation velocity $v_{sat(n)}$ or $v_{sat(p)}$. These range near $1 \times 10^7$ cm/s and thus already reach the order of magnitude of the mean thermal velocity. Fields in the range up to typically $2 \times 10^5$ V/cm appear in space charge regions during blocking of devices. Hence for a wide range of blocking voltages, carriers in the space charge region move with saturation velocity.

Relationship (2.39) has been verified experimentally in pure silicon [Jac77], for doped silicon experimental data of the high-field drift velocity are not available. Using the doping dependent mobility $\mu_{n,p}^{(0)}(N)$ in (2.39), however, a physically reasonable dependence on $\mathsf{E}$ and doping density $N$ is obtained which satisfies general scaling requirements [Tho80]. The saturation velocities $v_{sat(n,p)}$ are assumed to be independent of $N$. An alternative approach for $v_{n,p}$ as functions of $\mathsf{E}$ and N has been given by Scharfetter and Gummel [Scf69], see Appendix A1.

[3] In [Jac77] a second expression for the saturation velocity of electrons has been given, it reads:

$$v_{sat(n)} = \frac{2.4 \cdot 10^7 \mathrm{cm/s}}{1 + 0.8 \cdot \exp(T/600)}.$$

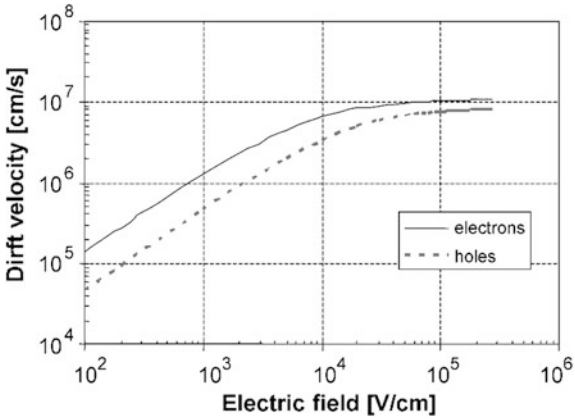



**Fig. 2.15** Drift velocity of electrons and holes as function of the electric field. Temperature 300 K

### 2.6.3 *Diffusion of Carriers, Current Transport Equations and Einstein Relation*

Unlike the situation in metals, the current in semiconductor devices is caused often not only by an electric field but also by diffusion of carriers. Generally, if some mobile particles have a spatially variable concentration C, they diffuse from a region of high to a region of low concentration. The particle current density J thus arising is proportional to the negative concentration gradient (Fick's first law):

$$J = -D \cdot \nabla C \tag{2.40}$$

where the proportionality factor $D$ is the diffusion constant. This holds also for electrons and holes, whose concentrations can vary because of a variation of the doping concentration or as result of injection of carriers. Due to the charge of the carriers the particle currents are connected with electrical currents. Multiplying with $\pm$ q and assuming the concentration gradient to appear in x direction, the electrical diffusion current densities are

$$j_{n,diff} = -q \cdot D_n \cdot \frac{dn}{dx} \tag{2.41}$$

$$j_{p,diff} = -q \cdot D_p \cdot \frac{dp}{dx} \tag{2.42}$$

Together with the field currents, Eq. (2.31), the total current densities are given by the transport equations:

$$j_n = q \cdot \left( \mu_n \cdot n \cdot \mathsf{E} + D_n \cdot \frac{dn}{dx} \right) \tag{2.43a}$$

$$j_p = q \cdot \left( \mu_p \cdot p \cdot \mathsf{E} - D_p \cdot \frac{dp}{dx} \right) \tag{2.43b}$$

The diffusion constants $D_n$, $D_p$ depend on the same scattering mechanisms as the mobilities. In fact, they are related to the mobilities by the following simple relationship

$$D_{n,p} = \frac{kT}{q} \cdot \mu_{n,p} \tag{2.44}$$

which is called *Einstein relation*. This can be derived from the case of thermal equilibrium [Sho59]: As follows from (2.43b) with the thermal equilibrium condition $j_p = 0$, a concentration gradient $dp/dx$ is connected with a field $\mathsf{E}$ and hence with an electrical potential $V(x) = -\int \mathsf{E} dx$. Now the holes obey the Boltzmann distribution

$$p(x) \propto \exp\left( -\frac{qV(x)}{kT} \right) \tag{2.45}$$

as can be concluded from (2.3). The energy states are located here at different points in space. On the other hand, one obtains from (2.43b) with $j_p = 0$

$$\begin{aligned} \frac{d \ln p}{dx} &= \frac{\mu_p}{D_p} \mathsf{E} = -\frac{\mu_p}{D_p} \cdot \frac{dV}{dx} \\ p &\propto \exp\left( -\frac{\mu_p}{D_p} V(x) \right) \end{aligned} \tag{2.46}$$

This is compatible with the Boltzmann distribution (2.45) only if $D_p/\mu_p = kT/q$. Similarly one can proceed for electrons. Hence the Einstein relation (2.45) is an immediate consequence of the Boltzmann distribution. The factor $kT/q$ in (2.45) has the dimension of a voltage and is called thermal voltage. Its value at 300 K is 25.85 mV. Hence the diffusion constants measured in $cm^2/s$ amount to only about 1/40th of the respective mobility in $cm^2/Vs$.

The Einstein relation is used very extensively not only for thermal equilibrium assumed for its derivation, but also for non-equilibrium up to high current densities and high injection levels. In a series of papers (see [Mna87a, Mna87b, Kan93, Mna98]) a model has been developed and applied claiming that the opposite drift velocities of electrons and holes in the on-state at high injection levels leads to complete breakdown of the Einstein relation due to a mutual drag. This model contradicts partly the conventional mobility theory and results in considerably

different device characteristics.[4] Hence the question arises which of the two approaches is appropriate for power device simulation.
The answer follows from the fact that virtually in all cases of interest in this respect the drift velocity of carriers is small against their thermal velocities:

$$v_{n,p} \ll v_{n,ptherm}. \tag{2.47}$$

If for example the electron current density $j_n$ in the base region of a pin-diode is as high as $5 \times 10^3 A/\mathrm{cm}^2$ and the electron concentration at its minimum $5 \times 10^{17}\mathrm{cm}^{-3}$, the drift velocity of electrons, $v_n = j_n/qn$, is still less than 1% of their mean thermal velocity at 300 K. In the whole range of operating temperatures and even down to 100 K the condition (2.47) is well fulfilled up to high current densities. At high fields, where (2.47) is not satisfied, the diffusion currents are usually negligible and the diffusion constants hence not relevant. Since the Maxwell distribution of carrier velocities holds still approximately, the microscopic scattering processes determining mobilities and diffusion constants are the same as in thermal equilibrium. Hence the relation between these constants should not be changed. Similarly as the doping density the concentrations of injected carriers determines the number of active and passive scatterers without influencing the relation between $\mu_{n,p}$ and $D_{n,p}$.

To prove this analogously to the thermal equilibrium case one has to take into account, that the mutual compensation of field and diffusion currents used in the above derivation does no more happen approximately at high current densities. One can assume however, regarding the case of holes, a field strength $\mathsf{E}_{eq,p} \equiv D_p/\mu_p d \ln p/dx$ (equilibrium field for holes), at which the hole diffusion current is compensated by the field current, the ohmic part of the field which is proportional to the hole current being disregarded. Depending on the injected carrier distribution the field $\mathsf{E}_{\mathrm{eq,p}}$ varies with space coordinate $x$. Since $j_p = 0$ for $\mathsf{E} = \mathsf{E}_{\mathrm{eq,p}}$, the Eq. (2.46) follows as above with the potential $V(x)$ resulting from this field. Because due to the condition (2.47) also the Boltzmann distribution (2.45) is valid approximately, the Einstein relation between $\mu_p$ and $D_p$ results for the field $\mathsf{E}_{eq,p}\ (x)$. Since (2.47) however is also the condition for field-independent mobilities (see Eqs. 2.30, 2.29), the relation (2.44) follows for the total field including the

[4]For example, the ambipolar diffusion constant

$$D = \frac{2D_nD_p}{D_n + D_p},$$

relevant for the carrier distribution at high injection levels (see Sect. 5.4.1), is proposed in mentioned papers to be independent of the carrier concentration $n = p$. While $D_n$ decreases with increasing $n$, the hole diffusion constant $D_p$ *rises*. Obviously this disagrees with (2.45) since $\mu_n$ *and* $\mu_p$ decrease with $n$. Also the mobility ratio $\mu_n/\mu_p$ is obtained to be constant in cited papers in contrast to the conventional theory. This results in strongly different high current characteristics of asymmetrical $\mathrm{pn}^-\mathrm{n}^+$ diodes.

ohmic part. The analogous argumentation applies to $\mu_n$ and $D_n$. Hence *under operating conditions of power devices including high current densities and high injection levels, the Einstein relation will be applicable with good accuracy.*

## 2.7 Recombination—Generation and Lifetime of Non-equilibrium Carriers

In thermal equilibrium, charge carriers are continuously generated thermally and they disappear with the same rate by recombination. In devices during operation, however, carrier densities in the active region are not in thermal equilibrium, they are higher or lower than the densities according to the equilibrium Eqs. (2.4), (2.5) and (2.6). A non-equilibrium state tends to restore itself to equilibrium. The time within which the system strives to achieve this, if injection/extraction and external generation are turned off, is determined by the *lifetime* $\tau$ of the non-equilibrium carriers. This is an adjustable quantity which is decisive for the dynamic as well as static behavior of power devices.

The definition of the lifetime uses the net recombination rates $R_n$ and $R_p$ of electrons and holes which are defined as the difference between the thermal recombination rates $r_{n,p}$ and thermal generation rates $g_{n,p}$:

$$R_n \equiv r_n - g_n, R_p \equiv r_p - g_p.$$

These thermodynamic quantities, which are zero in thermal equilibrium, describe the decrease of n and $p$ with time due to the net thermal recombination. Hence one has

$$R_n \equiv r_n - g_n = -\left(\frac{\partial n}{\partial t}\right)_{rg} \quad R_p \equiv r_p - g_p = -\left(\frac{\partial p}{\partial t}\right)_{rg} \tag{2.48}$$

where the index 'rg' marks the part of the time derivatives owing solely to recombination and generation. An in/outflow of carriers into/out of the considered volume element as well as external generation, for example by light, is excluded in (2.48). $R_n$, $R_p$ depend on doping and carrier densities and increase with increasing deviation of the respective carrier density from the equilibrium density $n_0$ or $p_0$. The relationship between $R_n$, $R_p$ and the excess concentration $\Delta n \equiv n - n_0$, respectively $\Delta p \equiv p - p_0$ is usually not very far from a linear dependency. Therefore it is useful to define lifetimes $\tau_n$, $\tau_p$ by the equations:

$$R_n \equiv \frac{\Delta n}{\tau_n}, \qquad R_p \equiv \frac{\Delta p}{\tau_p} \tag{2.49}$$

Compared with $R_n$, $R_p$, the lifetimes depend not very strongly on the respective excess concentration, and in some significant cases they are actually constant. This holds for the lifetime of injected minority carriers, the *minority carrier lifetime*, if the injection level is low (density of minority carriers is small compared to the majority concentration).
For the decay of a homogeneous excess concentration, e.g. of $\Delta p(t)$, Eqs. (2.48) and (2.49) yield:

$$\frac{d\Delta p}{dt} = -\frac{\Delta p}{\tau_p}, \qquad \Delta p = p(0) \cdot e^{-t/\tau_p} \tag{2.50}$$

where in the latter equation $\tau_p$ was assumed to be constant. From this the meaning of $\tau$ as a *lifetime* of the excess carriers becomes obvious. In the *stationary* case the disappearance of carriers due to recombination (for $R_n > 0$) is compensated by a net inflow of carriers or by external generation. In this case the net recombination rates of electrons and of holes are equal:

$$R_n = R_p = R \tag{2.51}$$

since the number of electrons leaving the conduction band must equal the number of electrons entering the valence band, the charge on possible intermediate levels being constant in the stationary case. If energy levels in the band gap are not involved, (2.51) is valid also for time dependent processes.

We have talked till now about recombination in the *volume* of the semiconductor. Before continuing with this, it is the point here to note that also *recombination at the metallic contacts*, especially at the anode contact, is very significant for modern power devices. The recombination at the anode metal layer contacting a p-region results in an electron particle current to the contact given by

$$J_n = (n - n_0)s \tag{2.52}$$

where s is the surface recombination velocity at the contact. For the recombination process the same hole current is needed, which flows in blocking direction and during on-state reduces the total hole current. It is used hence to control the emitter efficiency. s is typically of the order of magnitude $10^5$ cm/s and approximately constant. The high surface recombination velocity arises from a high density of interface states, distributed continuously over the bandgap. The recombination at a contact is controlled in devices by the integral doping concentration in front of it. Since this is treated later in the book together with the impact on switching properties (see Sect. 5.7.4.3), this note suffices here.

Returning to the *volume recombination* three physical *mechanisms* are to be mentioned: (i) recombination at recombination centers formed by ‘deep impurities’ or ‘traps’ which have energy levels deep in the band gap, (ii) band-to-band Auger recombination and (iii) radiative band-to-band recombination. The latter two mechanisms occur in the semiconductor lattice itself and depend only on carrier

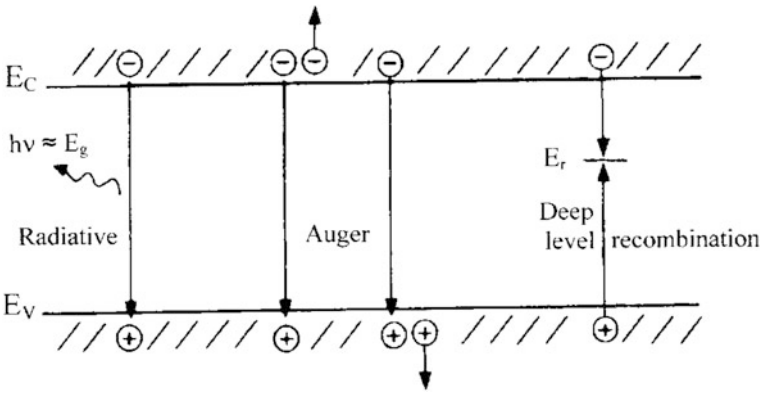



**Fig. 2.16** Mechanisms of recombination. **a** radiative band-to-band recombination. **b** Auger recombination with energy transfer to a third carrier and **c** recombination via a deep energy level

concentrations, not directly on the density of normal and deep impurities. The three mechanisms are illustrated in Fig. 2.16. The total recombination rate is made up additive of the single parts, hence according to (2.48) the inverse total *lifetime* is obtained by adding the reciprocal single lifetimes $\tau_i$ :

$$\frac{1}{\tau_{tot}} = \sum \frac{1}{\tau_i} \tag{2.53}$$

This applies also to the superposition of single lifetimes caused by different independent kinds of traps if present. In the following sections, the intrinsic mechanisms are described first.

### 2.7.1 Intrinsic Recombination Mechanisms

**(a) Radiative band-to-band recombination**: As shown in Sect. 2.4, this direct recombination of an electron and a hole under transfer of the released energy to a light quant has a high probability only in direct semiconductors. According to simple statistics the net recombination rate is

$$R = B \cdot (n \cdot p - n_i^2) \tag{2.54}$$

with the radiative recombination probability B.

The radiative lifetime, for example the hole lifetime in an n-type semiconductor, is obtained with (2.49) as $\tau_{prad} = \Delta p/R = 1/(B \cdot n)$ since $np - n_i^2 = n_0\Delta p + p_0\Delta n + \Delta n\Delta p = n\Delta p$ assuming $p_0 << n_0$: Hence, the radiative minority carrier lifetime is inversely proportional to the majority carrier density. In most

cases $n \cdot p >> n_i^2$, so that $R \approx B \cdot n \cdot p$. In GaAs the radiative lifetime is estimated to be 6 µs at a doping density N = $1 \times 10^{15}$ cm$^{-3}$ or 60 ns at $1 \times 10^{17}$ cm$^{-3}$ [Atk85]. This small lifetime limits the applicability of GaAs for bipolar devices. In silicon, the recombination constant at 300 K is B $\approx 1 \times 10^{-14}$ cm$^3$/s [Sco74] yielding $\tau_{rad}$ = 1 ms at a majority carrier density n = $1 \times 10^{17}$ cm$^{-3}$. In practice such a high lifetime is not measured in silicon devices, because the other recombination mechanisms are more effective (see the next sections). Using the connection between the injected carrier concentrations and the intensity of the recombination radiation, the radiation is used to investigate the inner operation of devices.

**(b) Band-to-band Auger recombination**: In Auger recombination, the energy released during the recombination event is transferred not to a light quant but to a third electron or hole, where participation of a phonon can be required for conservation of momentum. Therefore the recombination probability B in (2.54) has to be replaced by a factor which is proportional to the carrier concentrations. Hence the Auger recombination rate is

$$R_A = \left(c_{A,n} \cdot n + c_{A,p} \cdot p\right) \cdot \left(n \cdot p - n_i^2\right) \tag{2.55}$$

The coefficients $c_{A,n}$, $c_{A,p}$ determine the recombination rate for the cases that the third carrier taking the energy away is an electron, respectively a hole. Since the concentrations appear with power 3, the probability of this mechanism increases strongly with carrier concentration, and the lifetime *decreases* strongly. Therefore the Auger recombination is important mainly in highly doped regions. In an $n^+$ region with a small concentration of injected holes, with $p << n$ and $n \cdot p >> n_i^2$ (2.55) turns into $R_A = c_{A,n} \cdot n^2 \cdot p$, and for the hole lifetime one obtains from (2.49):

$$\tau_{A,p} = \frac{p}{R_A} = \frac{1}{c_{A,n} \cdot n^2} \tag{2.56}$$

where the extremely small equilibrium concentration $p_0$ has been neglected. The formula for the electron lifetime in a $p^+$ region is formed analogously. The Auger coefficients in silicon are in the range of $10^{-31}$ cm$^6$/s, according to [Dzi77] their values are:

$$c_{A,n} = 2.8 \times 10^{-31} \mathrm{cm}^6/s, \quad c_{A,p} = 1 \times 10^{-31}\ \mathrm{cm}^6/s \tag{2.57}$$

They are approximately independent of temperature. For a doping density of $1 \times 10^{19}$ cm$^{-3}$, the Auger electron lifetime in a $p^+$ region is $\tau_{A,n} = 1/(c_{A,p} \cdot p^2) = 0.1\ \mu s$, and the hole lifetime in an $n^+$ region is 0.036 µs. The small lifetime in highly doped regions is a constituent part of the h parameters via which the properties of these regions influence the characteristics of devices. This will be shown in Sect. 3.4.

Another case where the Auger recombination is of significance in devices is that of high concentrations of injected carriers in a weakly doped base region. With

neglect of the doping density the neutrality requires p ≈ n which inserted in (2.55) yields

$$R_{A,hl} = \left(c_{A,n} + c_{A,p}\right) \cdot p^3 \tag{2.58}$$

Hence, the high-level Auger lifetime is

$$\tau_{A,hl} = \frac{1}{\left(c_{A,n} + c_{A,p}\right) \cdot p^2} \tag{2.59}$$

At $p = n = 3 \times 10^{17}$ cm$^{-3}$, this relation together with (2.57) results in an Auger lifetime of 29 µs. At high current densities, the Auger recombination in the base region of high voltage devices becomes noticeable.

### 2.7.2 *Recombination at Recombination Centers Including Gold, Platinum and Radiation Defects*

The recombination via deep energy levels in the band gap caused by appropriate 'deep impurities' or lattice imperfections is the dominant mechanism of recombination in lowly and intermediately doped regions in silicon devices. By these recombination centers called also 'traps' the lifetime can be controlled over a wide range, which is commonly used to reduce the switching time and switching losses of devices for higher frequencies. The doping with deep impurities is carried out after the normal doping which determines the conductivity. In the history of device technology, at first gold was used as deep impurity for lifetime control in silicon. Meanwhile, many power devices are diffused with platinum, and most important is now the creation of lattice defects with deep levels by electron, proton or α-particle irradiation.

Unfortunately the reduction of switching time is associated with an increase of the on-state voltage drop and the high-temperature blocking current and often also

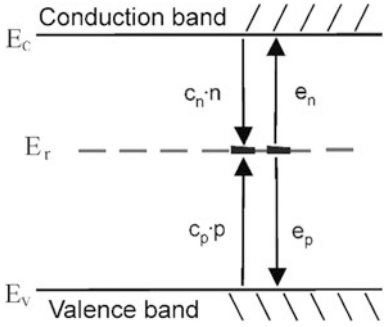



**Fig. 2.17** Capture and emission of carriers at a recombination center

of the resistivity of the base region by compensation. Since dynamic and stationary device properties are hence intimately connected with the adjustment of the lifetime, this is a central point in the development and fabrication of power devices. The recombination at a deep impurity proceeds in two steps, the capture of a conduction electron which then occupies the deep energy level, and thereafter a falling down of the electron into an empty place of the valence band, meaning capture of a hole by the impurity (see Fig. 2.17). Vice versa, an electron-hole pair is generated by thermal emission of a valence electron first to the impurity level, i.e. emission of a hole from the impurity to the valence band, and then emission of the electron from the impurity level to the conduction band.

The energy released during capture of a carrier is transferred to lattice vibrations, and the energy required conversely for generation is taken up from the lattice. Because of the large band-to-level distance a series of phonons are emitted respectively absorbed during capture and emission. These multi-phonon processes of capture and emission, however, are considered as a whole and described by over-all capture and emission probabilities. Using this concept Shockley, Read [Sho52] and Hall [Hal52] have developed a theoretical model (SRH model) which describes how recombination and generation depends on the level position, the capture and emission probabilities, type and concentration of normal doping, the injection level (minority carrier concentration) and temperature. This model is discussed in the following in detail. Although most deep impurities have two or even more levels, first centers with one level are considered.

**(a) Recombination centers with one deep level**: We consider the recombination at a center $R$ which can appear in a neutral and a negative charge state $R^0$, $R^-$. The impurity level is called in this case an acceptor level independent of its position in the bandgap. Similarly, if the charge state of the impurity atom changes from positive to neutral when the level is occupied by an electron, the level is called a donor level. The capture of electrons by centers $R^0$ causes an electron recombination rate $r_n = c_n \cdot n \cdot N_r^0$ where $N_r^0$ is the concentration of the neutral centers and $c_n$ a constant called capture probability or capture rate. The electron *generation* rate, given by emission of electrons from centers $R^-$ to the conduction band, is proportional to the concentration of negatively charged centers $N_r^-$, $g_n = e_n \cdot N_r^-$, where the constant $e_n$ is the emission probability also called emission rate of electrons. Hence the net recombination rate $R_n$ is

$$R_n = c_n n N_r^0 - e_n N_r^- \tag{2.60}$$

From thermal equilibrium whith $R_n = 0$, one obtains, indicating the concentrations by a '0' and using (2.21), (2.6), the following relationship between emission and capture probability

$$e_n = c_n \frac{n_0 N_{r0}^0}{N_{r0}^-} = c_n n_r \tag{2.61}$$

with

$$n_r = N_c g \exp\left(-\frac{E_c - E_r}{kT}\right) \tag{2.61a}$$

Similarly, the capture of holes by $R^-$ and emission (generation) of holes from $R^0$ result in the net recombination rate of holes

$$R_p = c_p p N_r^- - e_p N_r^0 \tag{2.62}$$

where $c_p$ is the hole capture probability and $e_p$ the emission probability of holes. A relationship between these quantities is obtained again from thermal equilibrium with $R_p = 0$ using (2.21)

$$e_p = c_p \frac{p_0 N_{r0}^-}{N_{r0}^0} = c_p p_r \tag{2.63}$$

with

$$p_r = \frac{N_v}{g} \exp\left(-\frac{E_r - E_v}{kT}\right) = n_i^2 / n_r \tag{2.63a}$$

The concentrations $n_r$, $p_r$ which relate the emission probabilities to the corresponding capture probabilities are apart from the degeneracy factor g identical with the electron or hole density, respectively, which are present if the Fermi level coincides with the recombination level. Using the relation $n_r p_r = n_i^2$, Eqs. (2.61), (2.63) yield:

$$e_n e_p = c_n c_p n_i^2 \tag{2.64}$$

Resulting from thermal equilibrium of elementary reactions of the recombination process, the relations between capture and emission rates are called detailed balance equations. Since the capture rates are defined in a low-field environment, also the emission rates correlated with them are low-field emission rates. Up to a field strength of $1 \times 10^5$ V/cm, however, they seem to be constant [LuN87].

With the definition of $n_r$, $p_r$ Eqs. (2.60), (2.62) take the form:

$$R_n = c_n (n N_r^0 - n_r N_r^-) \tag{2.65}$$

$$R_p = c_p (p N_r^- - p_r N_r^0) \tag{2.66}$$

As mentioned earlier, in the stationary case and generally if the time variation of charge on the deep level is negligible, one has $R_n = R_p$. Equating hence the right hand sides of (2.65) and (2.66) and using the total concentration $N_r = N_r^0 + N_r^-$, one can solve for $N_r^-$ and $N_r^0$ to obtain:

$$N_r^- = \frac{c_n n + c_p p_r}{c_n(n + n_r) + c_p(p + p_r)} N_r, \quad N_r^0 = N_r - N_r^- \tag{2.67}$$

Inserting this into Eqs. (2.65) and (2.66) the net recombination rate is obtained as the following function of $n$ and $p$:

$$\begin{aligned} R_n = R_p = R &= c_n c_p N_r \frac{n \cdot p - n_i^2}{c_n(n + n_r) + c_p(p + p_r)} \\ &= \frac{n \cdot p - n_i^2}{\tau_{p0} \cdot n + \tau_{n0} \cdot p + \tau_g \cdot n_i} \end{aligned} \tag{2.68}$$

$\tau_{n0}$, $\tau_{p0}$ and $\tau_g$ are lifetime quantities defined as

$$\tau_{p0} = \frac{1}{N_r c_p}, \quad \tau_{n0} = \frac{1}{N_r c_n} \tag{2.69}$$

$$\tau_g = \frac{n_i}{N_r}\left[\frac{1}{e_n} + \frac{1}{e_p}\right] \tag{2.70}$$

These are the central equations of the SRH model. They are valid also for a donor level except that $N_r^0$ has to be replaced by $N_r^+$ and $N_r^-$ by $N_r^0$ and the degeneracy factor g by 1/g. For the hole concentration $p_d = p_r$ of a donor level Eq. (2.63a) turns into Eq. (2.61a) with $N_c$ replaced by $N_v$, the equation for $p_d$ has the same form as the concentration $n_r = n_a$ of an acceptor level. The capture coefficients have typically values in the range $10^{-9}$ to $3 \times 10^{-7}$ cm$^{-3}$/s at 300 K, they decrease mostly slightly with temperature. So, a concentration $N_r$ of only $1 \times 10^{13}$ cm$^{-3}$ results in lifetime values $\tau_{n0}$, $\tau_{p0}$ in the range 0.3–100 µs.

We consider first the consequences of the SRH model for the minority carrier lifetime in a neutral region, thereafter the generation rate in a space charge region. Together with the definition (2.49), Eq. (2.68) yields the lifetime as function of n and p. In an n-type region with $n_0 \gg p_0$ and $np - n_i^2 = n_0 \Delta p + p_0 \Delta n + \Delta n \Delta p$ the hole lifetime $\tau_p = \Delta p/R$ is obtained as follows:

$$\tau_p = \tau_{p0} + \tau_{n0}\frac{p}{n} + \tau_g \frac{n_i}{n} = \tau_{p0}\left(1 + \frac{n_r}{n}\right) + \tau_{n0}\left(\frac{p + p_r}{n}\right) \tag{2.71}$$

In the case of neutrality, the electron and hole concentrations are connected with each other by the condition $n = N_D^+ + p \approx n_0 + p$, if the charge on the traps can be neglected. This is often allowed in a first approximation because the trap

concentration is mostly about an order of magnitude smaller than that of the normal doping. According to Eqs. (2.61a) and (2.63a) $n_r$, $p_r$ depend exponentially on the position of the recombination level and on temperature. Except the level lies near the middle of the bandgap the concentration referring to the more distant band, $n_r$ or $p_r$, is negligible. At least at higher temperatures, the higher of the concentrations $n_r$, $p_r$ has a considerable effect on $\tau$ assuming the concentration $n_0$ is not much higher than about $10^{14}$ cm$^{-3}$. The lifetime at low injection level ($p << n_0 \approx n$) is

$$\tau_{p,LL} = \tau_{p0}(1 + \frac{n_r}{n_0}) + \tau_{n0}\frac{p_r}{n_0} \tag{2.72}$$

and this becomes equal to $\tau_{p0}$ if the recombination level is located near the middle of the gap or the temperature is low ($n_0 >> n_r, p_r$). Analogous holds for the electron lifetime $\tau_n$ in a p region. At high injection levels defined by the condition $n = p >> n_0, p_0$, both lifetimes turn into the high level lifetime

$$\tau_{hl} = \tau_{no} + \tau_{p0} + (\tau_{p0}n_r + \tau_{n0}p_r)/n \tag{2.73}$$

where also $\tau_{p0}n_r + \tau_{n0}p_r >> \tau_{n0}\, n_0$ was assumed. For a recombination level near a band edge and particularly at elevated temperatures, Eq. (2.73) showing a decrease of $\tau_{hl}$ with n applies to a considerable range of concentrations. As n increases further until also $n >> n_r, p_r$ the high-level lifetime approaches the limiting value

$$\tau_{HL} = \tau_{n0} + \tau_{p0} \tag{2.74}$$

The lifetime $\tau_{HL}$ is larger than the low-level lifetime if $\tau_{p0}n_r + \tau_{n0}p_r < \tau_{n0}n_0$ or

$$c_n/c_p \cdot n_r + p_r < \; n_0 \tag{2.75}$$

The dependency on the injected hole concentration is very simple if the approximate neutrality condition $p = n - n_0$ can be used. Inserting this in (2.71) it is seen that the variation of $\tau_p$ between the low and high injection values is monotonous. This holds also if the trap charge is taken into account in the neutrality condition. With the equilibrium electron concentration $n_0 = N_D - N_r/(1 + n_r/n_0)$ (assuming further an acceptor level) formula (2.75) is the general condition for an increase of the minority carrier lifetime in an n-region with increasing p. If the left hand side is *larger* than $n_0$, the lifetime decreases. As can be shown by insertion of $n = N_D + p - N_r^-$ into (2.67), also $N_r^-/N_r$ varies monotonously with $p$ between its equilibrium value $n_0/(n_0 + n_r)$ at $p = p_0$ and the high injection value $c_n/(c_n + c_p)$ at $p \gg n_0, n_r, p_r$.

Besides the dependency on the injection level also the temperature dependence is connected with the quantities $n_r$, $p_r$. If one of these is comparable or larger than $n_0$ the variation of $n_r$ or $p_r$ with $T$ results in a strong increase of $\tau$ with temperature. Furthermore a decrease with increasing doping concentration $N_D \approx n_0$ results. As follows from (2.72) the low-level lifetime decreases from

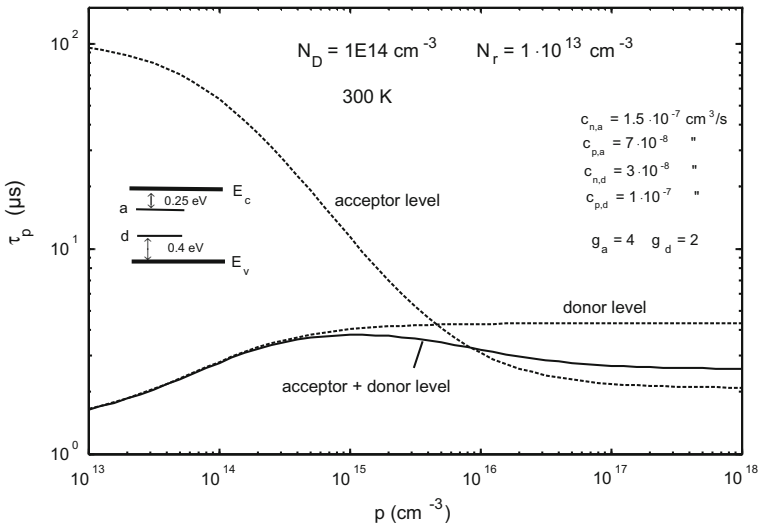



**Fig. 2.18** Hole lifetime in n-type silicon for three exemplary recombination centers, (1) a donor trap with a level 0.4 eV above the valence band, (2) an acceptor trap with a level 0.25 eV below the conduction band (dashed lines) and (3) a trap possessing both levels for itself (solid line)

$$\tau_p = \frac{\tau_{p0} n_r + \tau_{n0} p_r}{n_0} \quad \text{at} \quad n_0 \ll \max(n_r, p_r) \tag{2.76}$$

to $\tau_{p0}$ for $n_0 \gg n_r$, $p_r$. Under the condition max($n_r$, $p_r$) >> $n_0$ the lifetime $\tau_{p,LL}$, varies inversely proportional to $n_0$. By interchange of $n$ and $p$ the same expression results for the low-level electron lifetime in p-silicon except that $n_0$ is replaced by $p_0$. Hence the low-level minority carrier lifetime in $n$- and $p$-silicon with equal trap and doping density is equal under these conditions.

In Fig. 2.18 the lifetime $\tau_p(p)$ in n-Si is depicted for three examples, a donor trap with $E_r$ = 0.4 eV above the valence band, an acceptor center with $E_r$ = 0.25 eV below the conduction band and a center which possesses these two levels for its own (solid curve), which case will be discussed later. To distinguish between the capture rates for the donor and acceptor level they are indicated by an additional subscript 'd' respectively 'a'. The examples are chosen to illustrate how the lifetime depends on the position of the level in the bandgap. The capture probabilities are in the usual range. As degeneracy factors the usual values $g_a$ = 4 for the acceptor and $g_d$ = 2 for the donor level were used. The concentration of the recombination centers ($1 \times 10^{13}$ cm$^{-3}$) is significantly smaller than the normal doping density ($1 \times 10^{14}$ cm$^{-3}$). The charge on the recombination centers was taken into account in the calculation, but its effect is not large since $N_r << N_D$ and additionally at low injection levels the centers are mostly neutral because the Fermi level lies below the acceptor level and above the donor level. As is seen, the lifetime for the donor trap increases with p as is expected from (2.71) and (2.75), since $n_r$ is negligible and

according to Eq. (2.63a) (in the donor version, $g \rightarrow 1/g$) $p_r$ is with $1.18 \times 10^{13}$ cm$^{-3}$ smaller than $n_0$. The lifetime caused by the acceptor center on the other hand decreases from a very high value at small p to a low high-level lifetime. Also this follows immediately from (2.71) since $n_r = 7.22 \times 10^{15}$ cm$^{-3}$ >> $n_0$. At higher temperatures not included in the figure the lifetime particularly of the acceptor impurity is still much higher at small p and the decrease with increasing p correspondingly stronger. Similar temperature dependences are shown further below in the case of platinum.

What is the cause of the high low-level lifetime of the center with the level near the conduction band ($n_r >> n_0$)? The capture and emission rates of electrons are still nearly in thermal equilibrium, which means $R_n \approx 0$. With (2.65) this results in $N_r^- = n_0/n_r \cdot N_r^0 \approx n_0/n_r \cdot N_r << N_r$. Only very few centers are hence in the negative charge state in which they can capture a hole. Inserting $N_r^-$ into Eq. (2.66), whose hole emission term is negligible against the capture term for $p \geq p_0$, one obtains $\tau_{p,LL} = 1/\left(c_p N_r^-\right) \approx \tau_{p0} \cdot n_r/n_0 \gg \tau_{p0}$. Also if the level lies near the valence band ($p_r >> n_0$), the lifetime $\tau_{p,LL}$ is very high according to (2.72). In this case the capture and emission rate of *holes* are approximately in equilibrium, hence it follows from (2.66) $N_r^0 = p/p_r N_r$. Inserting this into (2.65) and neglecting the term with $n_r$, one obtains $R_p = R_n = c_n np/p_r \cdot N_r$ and together with (2.50) $\tau_{p,LL} = 1/(c_n N_r) \cdot p_r/n_0$. The *hole* lifetime is controlled here by the capture of *electrons* by the few $R^0$ atoms, which leaves the high number of capture and emission events of holes nearly in equilibrium. The general result is that an energy level near a band edge is little effective in recombination at low doping and injection levels with $n << n_r$ or $n << p_r$.

Both the very low ratio of high-level to low-level lifetime and the strong increase of the lifetime with temperature obtained for a near-band level are unfavorable for devices. The first property worsens the relation between on-state voltage drop and switching and recovery times. For thyristor's and IGBTs it is additionally unfavorable because the high low-level lifetime results in a high amplification of the generation current in the space charge region by the p$^+$np partial transistor [Cor74, Bal77]. In both cases the high-level to low-level lifetime ratio is particularly crucial at elevated temperatures, so the high *temperature* dependence comes into play here. A strong increase of the lifetime with temperature even at high injection levels can lead, furthermore, to a strong decrease of the on-state voltage with temperature, and this makes devices unsuitable for parallel connection [Lut00, Sie01] (see Sect. 5.4.6). *Hence a recombination level near a band edge is disadvantageous*. It can be made innocuous, however, by other levels, which lie nearer to the middle of the gap.

Equation (2.71) is also valid if the equilibrium is undercut, i.e. if $np < n_i^2$, provided $n >> p_0$. In a **space charge region** as found in reverse-biased pn junctions, however, usually *both* carrier concentrations are negligible. With $n = p = 0$ the generation rate is obtained directly from (2.68), (2.70) as:

$$-R = G = \frac{n_i}{\tau_g} = \frac{N_r}{1/e_n + 1/e_p} \tag{2.77}$$

Determining the generation, the time quantity $\tau_g$ is called a generation lifetime. Since it does not obey (2.49), however, it is strictly speaking not a carrier lifetime in the usual sense. $\tau_g$ and hence G depend strongly on the position of the level in the bandgap. Owing to the inverse interrelation between $e_n$ and $e_p$ (see Eq. 2.64), $\tau_g$ reaches its minimum and G its maximum at that energy $E_r = E_{r,m}$ where $e_n = e_p = \sqrt{c_n c_p}\, n_i$. Inserting this into (2.70) the generation lifetime at its minimum is obtained as $\tau_{g\,\min} = 2/(N_r\sqrt{c_n c_p}) = 2\sqrt{\tau_{n0}\tau_{p0}}$: the minimal value of $\tau_g$ is twice the geometrical mean value of $\tau_{n0}$ and $\tau_{p0}$. The energy of maximal G is given by

$$E_{rm} = (E_c + E_v)/2 + kT/2 \cdot \ln(c_p N_v/(c_n N_c g^2))$$

which is not far from the bandgap middle. The generation rate can be written $G = G_{\max}/\cosh((E_r - E_{rm})/kT)$. A few $kT$ away from $E_{r,m}$, G is proportional to the smaller one of the emission rates and hence decreases exponentially with increasing distance of the level from the more distant band. If short switching times of a device are required, but on the other side also a small generation in the space charge region and hence low blocking current, this can be reached by choosing a deep impurity whose recombination level is well distant from the middle of the bandgap, but not *too* close to one of the band edges.

Although the density of recombination centers is mostly significantly smaller than the doping of the weakly doped base region, this may be so only to a less extent in very fast devices. Here the charge on the traps can have undesirable effects, such as a compensation of the normal doping (reduction of the conductivity), a reduction of the breakdown voltage or a premature punch-through of the space charge region. The influence of the traps on the carrier concentration in thermal equilibrium depends upon the position of their level relative to the Fermi energy and likewise on the type of the level. For example an acceptor-like impurity in a neutral n base with a level a few kT above the Fermi energy will be neutral to a large part and hence will not affect the free electron concentration essentially. A donor trap with a level at the same position on the other hand is positively charged and leads to an *increase* of the electron concentration in a neutral n region. In a space charge region of a reverse biased pn-junction, the charge in steady state is given by the condition that the generation rate of electrons must equal to that of holes. For an acceptor level one has $e_n N_r^- = e_p N_r^0$, hence $N_r^-/N_r^0 = e_p/e_n$. To obtain a low concentration of charged deep acceptors (e.g. to avoid punch-through), a higher emission rate $e_n$ than $e_p$ is required. Generally, the density of charged traps in a stationary non-equilibrium state is given by (2.67).

During switching processes, the densities $N_r^-, N_r^0$ follow the variation of n and p not instantaneously but often rather slowly. Their time-dependence is described by the equation

$$\frac{dN_r^-}{dt} = c_n n N_r^0 - c_p p N_r^- - e_n N_r^- + e_p N_r^0 = -\frac{dN_r^0}{dt} \tag{2.78}$$

which follows immediately from the capture and emission events recharging the impurity. Together with differential equations for n and p following from (2.60), (2.62) and (2.48) the charged trap density as a function of time can be numerically calculated. Some effects of the temporary charge on deep impurities will come up in Sect. 13.3.

In above examples the distances of the acceptor level from the conduction band and the donor level from the valence band have been assumed constant calculating the concentrations $n_a$ and $p_d$. Since the bandgap decreases with increasing temperature, at least one of the activation energies of a level, $\Delta E_n = E_c - E_r$ or $\Delta E_p = E_r - E_v$, varies also, mostly however both vary with T. This has a strong effect on the recombination. The variation of $\Delta E$ manifests itself in an 'effective' degeneracy factor $g'$ which usually strongly deviates from the values 2 or 4 of the 'intrinsic' degeneracy factor g introduced in Sect. 2.5. As pointed out in Appendix B, the concentrations $n_r$, $p_r$ can be calculated replacing in (2.61a) respectively (2.63a) the real activation energy $\Delta E(T)$ by a constant apparent experimental activation energy $\Delta E'$ and the degeneracy factor g by the effective degeneracy factor g′. In contrast to $g$, *the degeneracy factor* $g'_n$ *for* $n_r$ *is different from the degeneracy factor* $g'_p$ *of* $p_r$. Written in full the concentration $n_r = n_a$ of an acceptor level and $p_r = p_d$ of a donor level can be calculated from

$$n_a = N_c g'_{n,a} \exp(-\Delta E'_{n,a}/kT) \tag{2.79a}$$

$$p_d = N_v g'_{p,d} \exp(-\Delta E'_{p,d}/kT) \tag{2.79b}$$

$p_a$ and $n_d$ are then given as $n_i^2/n_a$ respectively $n_i^2/p_d$. In Appendix B it is shown how the degeneracy factors $g'$ are made up and can be evaluated from the variation of a level position with temperature.

Actually all deep impurities in silicon used for lifetime reduction possess more than one level in the bandgap. The SRH model for two levels and its application to gold and platinum will be treated in following sections in some detail. A part of the exposition is possibly helpful also for a better understanding of recombination centers introduced by radiation.

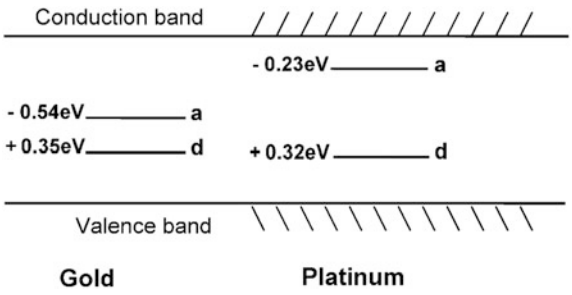



**Fig. 2.19** Energy levels of gold and platinum in silicon. Numbers with a plus sign indicate the distance of the level from the valence band edge, numbers with a minus sign the distance from the conduction band. Donor levels are labeled with "d" and acceptor levels with "a". The specified values neglect their temperature dependence

**(b) Recombination centers with two levels**: Au, Pt and several other deep impurities in Si possess a donor level in the lower half of the bandgap and an acceptor level in the upper half [Pal74] (see Fig. 2.19). The two levels arise because these centers have three charge states, whose neighboring ones can pass into one another. The effective concentrations of the levels are coupled with another and vary with Fermi level, injection level and temperature. Equations describing this have been developed in several papers [Sah58, Kon59, Scz66, Abb84]. In the present section a general lifetime equation for two-level traps is introduced, a numerical example and significant limiting cases are discussed. The notation is adapted to Au and Pt, centers with a donor and an acceptor level.

Recombination centers of this type can be positively charged ($R^+$), neutral ($R^0$) or negatively charged ($R^-$). According to the definition given above, the donor level indicates the energies involved in transitions between a band and the impurity if the charge state changes between $R^+$ in $R^0$, the acceptor level the energies between the bands and the impurity for changes between $R^0$ and $R^-$. As in the case of one level, the transition can occur in each case by capture or emission of an electron from resp. to the conduction band and a similar exchange of a hole with the valence band (see Fig. 2.17). Because of the electrostatic energy gained during the capture of a conduction electron by $R^+$ but not by $R^0$, the donor level lies (generally) below the acceptor level. This differs from the familiar position of donor and acceptor levels of normal doping atoms (see Fig. 2.5).

To obtain a formula for the lifetime the total recombination rate is expressed as the sum of two SRH terms (2.68), one for each level. The trap density $N_r$ is to be replaced in both cases by the concentration of occupied plus unoccupied states, hence for the donor level by $N_d = N_r^+ + N_r^0$ and for the acceptor level by $N_a = N_r^0 + N_r^-$. These concentrations are coupled via the constant total concentration $N_r = N_r^+ + N_r^0 + N_r^-$ and depend on n and p as is obtained in steady state setting $R_n = R_p$ for each level. The formulas can be found widely in [Kon59, Scz66, Abb84]. From the total recombination rate and the definition Eq. (2.49) one gets for the hole lifetime in an n-type semiconductor with $n_0 >> p_0$:

$$\tau_p^{(n-Si)} = \frac{1}{N_r c_{p,a}} \frac{\left(n + n_a + \frac{c_{p,a}}{c_{n,a}}(p_a + p)\right)\left(n + \frac{c_{p,d}}{c_{n,d}} p_d\right) + \left(n_a + \frac{c_{p,a}}{c_{n,a}} p\right)\left(n_d + \frac{c_{p,d}}{c_{n,d}} p\right)}{n c_{p,d}\left(\frac{n}{c_{p,d}} + \frac{n_a}{c_{p,a}} + \frac{p}{c_{n,a}} + \frac{p_d}{c_{n,d}}\right)} \tag{2.80}$$

$n_a$ and $p_a$ denote the concentrations $n_r$, $p_r$ of the acceptor level introduced by Eqs. (2.61a), (2.63a), $p_d$ and $n_d$ the concentrations $p_r$, $n_r$ of the donor level defined by these equations in the donor version where $N_{r0}^-$, $N_{r0}^0$ have to be replaced by $N_{r0}^0$ and $N_{r0}^+$, respectively, and g by 1/g. Actually, Eq. (2.79a, b) will be used often to calculate $n_a$, and $p_d$ from the constant apparent activation energies $\Delta E'_{n,a}$, $\Delta E'_{p,d}$ and the effective degeneracy factors $g'_{n,a}$, $g'_{p,d}$. Mostly, if the acceptor level is well in the

upper half and the donor level in the lower half of the bandgap, the concentrations $p_a$ and $n_d$ are negligible. The electron concentration in a neutral n region is given by the hole density according to $n = N_D + p + N_r^+ - N_r^-$, where the concentration $N_r^+ - N_r^-$ representing the charge of the traps can be neglected often in a first approximation. In our calculations it is taken into account using formula (B17) given in Appendix B.2. The electron lifetime in a p-type region follows from (2.80) by interchanging n and p in letters and indices as well as indices a and d.

In Fig. 2.18 the lifetime following from (2.80) for a trap with the shown levels and capture rates is plotted as solid curve. The distances of the donor level from the valence band and of the acceptor level from the conduction band were assumed temperature independent. The same capture rates and degeneracy factors are used as for the above discussed traps which have only one of the two levels (dashed curves). As is seen the lifetime of the two-level center is up to about $p = 4\ 10^{14}\ \mathrm{cm}^{-3}$ the same as if only the donor level would be present. The lifetime is strongly reduced in this range by the donor level. Responsible for this are the equilibrium concentration $n_0$ ($\approx 9 \times 10^{13}\ \mathrm{cm}^{-3}$) and the concentrations $p_d$ ($1.18 \times 10^{13}\ \mathrm{cm}^{-3}$) and $n_a$ which latter with $7.22 \times 10^{15}\ \mathrm{cm}^{-3}$ is two orders of magnitude higher than $p_d$ and $n_0$. Under these conditions and for relatively low injection levels with $n_0 + p << n_a$ Eq. (2.80) reduces to

$$\tau_p = \frac{1}{N_r}\left(\frac{1}{c_{p,d}} + \frac{1}{c_{n,d}}\frac{p_d + p}{n}\right) \quad (\text{for} \quad n, p_d \ll n_a) \tag{2.81}$$

In the precondition for this equationratios of capture coefficients have been ignored (set equal to 1) for simplicity. In words we have obtained: If the donor level is more distant from the valence band than the acceptor level from the conduction band and the Fermi level in thermal equilibrium below the acceptor level, the hole lifetime in the range $n_0 + p << \mathrm{n}_a$ is determined by the donor level. Equation (2.81) is identical with (2.71) with $n_r$ neglected and $p_r \equiv p_d$. The unfavorable consequences of the near band position of the acceptor level are eliminated by the donor level. The trap atoms are mostly in a charge state (neutral) in which they cannot react via the acceptor level with minority carriers.

If with increasing injection level the precondition $\mathrm{n} << n_a$ of (2.81) becomes invalid, the acceptor level too influences the lifetime. Contrary to the case of one level this can lead to a maximum of the $\tau_p(p)$-function as shown by Fig. 2.18. The limiting value of the high-level lifetime for $p >> n_0$, $p_0$, $n_a$, $p_d$ is obtained from (2.80) as

$$\tau_{HL} = \frac{1}{N_r} \cdot \frac{c_{n,a}/c_{p,a} + 1 + c_{p,d}/c_{n,d}}{c_{n,a} + c_{p,d}} \tag{2.82}$$

In the case of Fig. 2.18 $\tau_{HL}$ is higher than for the acceptor level alone although both levels contribute to it.

A simple limiting case appears if the Fermi level is well above the acceptor level so that $n_0 \gg n_a$. Then the low-level hole lifetime is obtained from (2.80) as

$\tau_{p,LL} = 1/\left(N_r c_{p,a}\right) \equiv \tau_{p0}^{(a)}$. Analogously, the low-level electron lifetime in p-silicon with $p_0 >> p_d$ is given by $\tau_{n,LL}^{(p-Si)} = 1/\left(N_r c_{n,d}\right) \equiv \tau_{n0}^{(d)}$ as follows from the equation for $\tau_n$ in p-Si obtained from (2.80) in the mentioned way. These cases are used to determine the capture probabilities $c_{p,a}$ and $c_{n,d}$.

Finally we consider a few limiting cases which we will meet again below. If under the condition $n_a >> p_d$ also the concentration $n_0$ is large against $p_d$, Eq. (2.80) yields for the low-level hole lifetime in an n-region:

$$\tau_{p,LL} = \frac{1}{N_r} \frac{1 + n_a/n_0}{c_{p,a} + c_{p,d} n_a/n_0} \quad (\text{for } \mathrm{n_0}, \mathrm{n_a} \gg \mathrm{p_d}) \tag{2.83}$$

Here the low-level lifetime is influenced by both levels. Compared with (2.81) the temperature dependence is weaker, because the denominator counteracts the T-dependence of the nominator and, furthermore, the concentration $n_a$ is less temperature dependent than $p_d$ due to the smaller distance from the relevant band.

If on the other hand $n_a << p_d$, as is the case of gold, the low-level minority carrier lifetime is obtained from Eqs. (2.79a, b) *for arbitrary doping concentrations* as

$$\tau_{p,LL} = \frac{1}{N_r c_{p,a}} \left(1 + \frac{n_a}{n_0}\right) \tag{2.84}$$

where the precondition is strictly speaking $n_a \ll c_{p,a}/c_{n,d}\, p_d$. Equation (2.84) is identical with Eq. (2.72) applied to the acceptor level.

At higher temperatures and low doping both $p_d$ and $n_a$ are large compared with the equilibrium concentration $n_0$. Then one obtains from Eq. (2.80) for the low-level hole lifetime in n-Si:

$$N_r \tau_{p,LL} = \left\{ \left(c_{p,a}/n_a + c_{n,d}/p_d\right) n_0 \right\}^{-1} \quad \text{for } n_a, p_d \gg n_0 \tag{2.85}$$

This is the extension of (2.76) to the present two-level case. The lifetime in this range is inversely proportional to $n_0$. The same expression, with $n_0$ replaced by $p_0$, holds for the minority carrier lifetime in p-silicon. Hence as for one level the low-level minority carrier lifetimes in n and p silicon are equal, if doping as well as trap concentrations are equal in both cases.

**(c) The recombination center gold**: Gold is the most investigated deep impurity in silicon, although disagreements between the parameter values obtained remain. The two-level model as due to three charge states of the same impurity has been called into question [Vec76, Lan80], but was reconfirmed by other authors [LuN87]. The levels are shown in Fig. 2.19 with rough numerical positions neglecting the temperature dependence. The acceptor level is located only a little above the middle of the bandgap, which has the obvious drawback of causing a high generation current in reverse biased junctions. Although gold has lost therefore a part of its

applications to platinum and radiation induced traps, it is still applied, because in other respects it has advantages. Gold attracts attention also as a kind of model trap.

Since according to the position of the levels $n_a << p_d$, the low-level hole lifetime in n-Si is given by Eq. (2.84). Hence *the low-level minority carrier lifetime in n-Si as function of doping density and temperature is determined by the acceptor level*. For *arbitrary* injection levels one obtains from Eq. (2.80) using $n_a \ll p_d$

$$\tau_p = \frac{1}{N_r c_{p,a}} \frac{\left(n + n_a + \frac{c_{p,a}}{c_{n,a}} p\right)\left(n + \frac{c_{p,d}}{c_{n,d}} p_d\right) + \left(n_a + \frac{c_{p,a}}{c_{n,a}} p\right) \frac{c_{p,d}}{c_{n,d}} p}{n\, c_{p,d}\left(\frac{n}{c_{p,d}} + \frac{p_d}{c_{n,d}} + \frac{p}{c_{n,a}}\right)} \tag{2.86}$$

where also $n_d$ and furthermore $c_{p,a}/c_{n,a}\, p_a$ against $n_a$ are neglected. Although the acceptor level lies near the middle of the bandgap, the latter neglect is justified up to 450 K, because the degeneracy factor $g'_{n,a}$ of the acceptor level turns out to be high.

A complete set of capture rates have been determined by Fairfield and Gokhale [Fai65] and later by Wu and Peaker [WuP82] who determined also their temperature dependences and listed results published till then. Because these and later results scatter strongly and are not sufficiently in agreement with lifetime measurements, we adjust the parameters first as far as necessary to the following lifetime properties of gold:

I. The hole lifetime in an n region at room temperature increases between low and high injection level by a factor of about 5. In [Fai65] the ratio of high-level to low-level was obtained to be 6.3, Zimmermann measured a ratio 4.0 [Zim73] and Hangleiter 5.5 [Han87]. The doping concentration in [Zim73] was $1 \times 10^{14}$ cm$^{-3}$, in [Fai65] and [Han87] higher doped devices with $N_D >> p_d = 1.31 \times 10^{14}$ cm$^{-3}$ were used.
II. The electron lifetime in a p-region with a doping concentration $N_A >>$ p$_d$ is independent of the injection level [Han87].
III. Both the high-level lifetime $\tau_{HL}$ and the low-level electron lifetime in a p-region with $N_A \gg p_d$ are proportional to $T^2$ over a wide range ($\tau_{n,LL}$ from 100 to 300 K [Scm82], $\tau_{HL}$ from 100 to 400 K [Sco76] (approximately confirmed in [Han87]).
IV. The electron emission rate of the gold acceptor level is an order of magnitude higher than the hole emission rate [Sah69, Eng75]. According to the detailed balance Eqs. (2.61), (2.63) this has to be taken into account for the choice of capture probabilities and degeneracy factor.

The capture rates of [Fai65] satisfy criterion I, but not II, the data of [WuP82] result in an unrealistic high-level to low-level ratio of 25.6. As is explained in detail in Appendix B, the above criteria together with published work lead us to the capture rates and degeneracy factors shown in Table 2.3. The 300 K-value of $c_{p,a}$ is adopted from [Fai65]. Using this and the emission rates of Sah et al. [Sah69] the value of $c_{n,a}$ is determined from the detailed balance Eq. (2.64). In agreement with [WuP82] the capture rates $c_{n,a}$ and $c_{n,d}$ are well an order of magnitude smaller than

the corresponding capture probabilities of holes. Hence according to Eq. (2.82) the high-level lifetime is determined essentially by the capture probability $c_{n,d}$ of the donor level. The increase of the lifetime between low and high level by a factor 5 to 6 quoted in criterion I, requires hence that $c_{n,d}$ is by about the same factor smaller than $c_{p,a}$, the capture rate determining the low-level lifetime in n-Si at 300 K. The temperature dependences are taken from literature considering criterion III. According to Eq. (B9) the effective degeneracy factor $g'_{n,a} = 10.9$ means, that the distance of the acceptor level from the conduction band decreases with temperature with the rate $0.795 \times 10^{-4}$ eV/K. This is 29% of the rate of the bandgap decrease at 350 K, if calculated from Eq. (2.9) together with the parameters of Table 2.1.

**Table 2.3** Capture probabilities, degeneracy factors and characteristic concentrations of the gold recombination center

| Acceptor level | Donor level | Unit |
|---|---|---|
| $c_{n,a} = 0.56(T/300\,\mathrm{K})^{0.5}$ | $c_{n,d} = 2.32(T/300\,\mathrm{K})^{-2}$ | $10^{-8}$ cm$^3$/s |
| $c_{p,a} = 11.5(T/300\,\mathrm{K})^{-1.7}$ | $c_{p,d} = 28.0(T/300\,\mathrm{K})^{0.5}$ | $10^{-8}$ cm$^{-3}$/s |
| $\Delta E'_{n,a} = 0.5472$ | $\Delta E'_{p,d} = 0.3450$ | eV |
| $g'_{n,a} = 10.9$ | $g'_{p,d} = 2.65$ | |
| $n_a(300\,\mathrm{K}) = 2.00 \times 10^{11}$ | $p_d(300\,\mathrm{K}) = 1.31 \times 10^{14}$ | cm$^{-3}$ |
| $n_\{a\}(400\,\mathrm{K}) = 6.26 \times 10^{13}$ | $p_d(400\,\mathrm{K}) = 6.29 \times 10^{15}$ | cm$^{-3}$ |

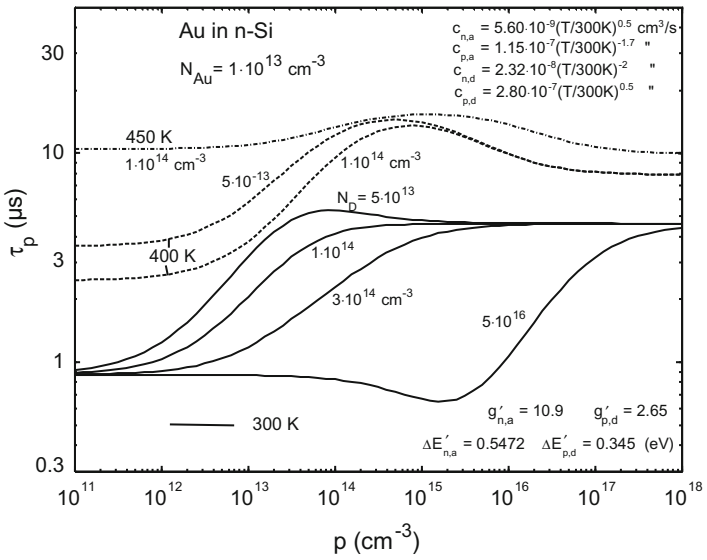



**Fig. 2.20** Hole lifetime for the gold recombination center in n-type silicon in dependence of the injected hole concentration. Parameters are the doping concentration and temperature

With $g'_{p,d} = 2.65$ the distance of the donor level from the valence band is obtained to be nearly constant.
Using these data in Eq. (2.86) the calculated injection dependence of the hole lifetime in n-silicon is shown in Fig. 2.20 for several doping concentrations and temperatures. The dependence on doping concentration and temperature is strong. Owing to the choice of the capture parameters the lifetime at 300 K increases by a factor of about 5. Also the increase of the high-level lifetime from 300 to 450 K accords with the $T^2$ dependency (criterion III). The minimum of the $\tau_p(p)$ curve for $N_D = 5 \times 10^{16}$ cm$^{-3}$ is not indicated by measurements (see [Fai65, Han87]), but since only weak it was possible not observable. Because of the good over-all agreement with measurements the minimum seems to be a necessary consequence of the theoretical model together with the criteria used for adjustment.

Since the criteria are reproduced, we can have some trust also in the other results. In Fig. 2.20 the $\tau_p(p)$-functions at 400 and 450 K show a maximum, as already seen in the case of Fig. 2.18. The acceptor level alone would yield an increase of the lifetime over the whole range, not only at 300 but also at 400 and 450 K, since the condition (2.75) is well satisfied. The shown decrease after the maximum at about $p = 5 \times 10^{14}$ cm$^{-3}$ is caused hence by recombination via the donor level. As mentioned, the high-level lifetime is (nearly) solely determined by the donor level since $c_{n,a} \ll c_{p,a}$ and $c_{n,d} \ll c_{p,d}$, whereas the low-level lifetime by the acceptor level. The ratio of high-level to low-level lifetime for $N_D = 1 \times 10^{14}$ cm$^{-3}$ is 3.0 at 400 K and about 1 at 450 K. *That the low-level lifetime is even at elevated temperatures not higher than the high-level lifetime is a significant advantage of gold* [see the discussion in section a)]. Since the concentration $n_a$ in (2.84) is small up to nearly 400 K (see Table 2.3), the low-level lifetime shows a relatively weak temperature dependence compared for example with platinum (see section d).

The curves for $N_D = 5 \times 10^{16}$ and $1 \times 10^{14}$ cm$^{-3}$ show that in device regions with higher n-doping ($5 \times 10^{16}$ cm$^{-3}$) used as 'field stop layers' the lifetime is significantly smaller than in the weakly doped base region for a given absolute injection level (constant gold density assumed). From Eq. (B14) in Appendix B it follows that the small lifetime over a wide injection range follows from the high coefficient $c_{p,d}/c_{n,a}$ in the denominator, hence is caused by recombination via the donor level. Only at very low injection levels the lifetime at room temperature is independent of $N_D$ and given by $\tau_{p,LL} = 1/(c_{p,a}N_r)$.

Also the minority carrier **lifetime in p-Si** is of interest for power devices, for example to simulate the recombination behavior of p-regions in thyristors and IGBTs. As stated in criterion II, at doping concentrations $N_A \gg p_d$, the case of the p-base of thyristors, the lifetime $\tau_n^{(p-Si)}$ is independent of the injection level. Rewriting Eq. (2.80) into the $\tau_n^{(p-Si)}$ version and using that $c_{n,a} \ll (c_{p,a}, c_{p,d})$ and $c_{n,d} \ll c_{p,d}$ a simple approximation results under the condition $p_0 \gg c_{n,a}/c_{p,a}n_a$, which down to $N_A = 1 \times 10^{14}$ cm$^{-3}$ and up to 400 K is well satisfied. The electron lifetime as function of injection level is then approximately given by:

$$\tau_n^{(p-Si)} \approx \frac{1}{N_r} \frac{1 + p/p_d}{c_{n,a} + c_{n,d} p/p_d} \tag{2.87}$$

Because $c_{n,a} < c_{n,d}$, the lifetime in p-silicon decreases with increasing injection $n \approx p - p_0$. For $p = p_0$ Eq. (2.87) is the counterpart to (2.83). The temperature dependence is only moderate, since the two $p_d$ terms counteract each other. As is noted also, the low-level lifetime at a doping density of $1 \times 10^{14}$ cm$^{-3}$ is by a factor of about 9 higher than in n-silicon at room temperature, if the gold concentration is equal.

Of interest for devices are furthermore the concentrations of the different charge states under different conditions. In an *n region in thermal equilibrium* the gold atoms are completely in the negative charge state, because the Fermi level is located well above the acceptor level. The equilibrium free electron concentration is reduced to $n_0 = N_D - N_{Au}$. Hence gold leads to an increase of the resistivity (compensation). At *high injection levels* with $n = p \gg n_a, p_a, p_d$, however, one obtains from equations (B13) to (B15) in Appendix B.2:

$$N_{Au}^+ : N_{Au}^0 : N_{Au}^- = \frac{c_{p,d}}{c_{n,d}} : 1 : \frac{c_{n,a}}{c_{p,a}} = 12.1 : 1 : 0.049$$

where for the numerical values the capture rates at 300 K were used (see Table 2.3). Hence at high injection gold is largely in the positive charge state, $N_{Au}^+ \approx N_{Au}$. In a *space charge region* under reverse bias on the other hand the generation rate in steady state $G = G_a = e_{n,a} N_{Au}^- = e_{p,a} N_{Au}^0$ yields $N_{Au}^- = e_{p,a}/e_{n,a} N_{Au}^0 \ll N_{Au}^0 \approx N_{Au}$. Because $e_{p,a}$ is more than an order of magnitude smaller than $e_{n,a}$, the gold centers in a space charge region are mostly neutral. The positive trap charge at high injection becomes noticeable during switching of a fast pin diode from forward to reverse bias, the negative charge in the n base in equilibrium during switching from zero to reverse bias. In the space charge region the $Au^+$ and $Au^-$ ions turn into the neutral state by emission of a hole respectively an electron. Because of the long time constant $1/e_{n,a}$ ($\approx$ 1 ms at room temperature) the decay of the concentration $N_{Au}^-$ is rather slow.

**(d) The recombination center platinum**. The levels of platinum in silicon have been determined by a series of investigators [Con71, Pal74, Mil76a, Sue94]. With some spreading they obtain a donor level 0.32 eV above the valence band and an acceptor level 0.23 eV below the conduction band as shown in Fig. 2.19. Some authors found additional levels attributed partly to a second Pt site present to a certain share of the total concentration [Lis75, Evw76, Sue94]. An acceptor level at 0.42 eV above the valence band reported as the dominant Pt recombination level [Lis75] has not been observed however by later researchers [Bro82b]. Since accepted by most, the level scheme of Fig. 2.19 is assumed in what follows. The

**Table 2.4** Capture cross sections of platinum in silicon determined in [Con71] and [Bro82a]. Also the exponents n of the power law temperature dependences $\sigma_{ij} \sim T^n$ and the effective degeneracy factors of [Con71] are given

| | [Con71] (295 K) | | [Bro82a] (77 K) | |
|---|---|---|---|---|
| | $\sigma$ ($10^{-16}$ cm$^2$) | n | $\sigma$ ($10^{-16}$ cm$^2$) | n |
| $\sigma_{p,a}$ | 55 | −4.4 | 30 | 0 |
| $\sigma_{n,a}$ | 63 | 0 | 8 | 0 |
| $\sigma_{p,d}$ | 16.2 | 0 | 3 | 0 |
| $\sigma_{n,d}$ | 18.8 | −4.0 | 1 | – |
| $g'_{n,a}$ | 1 ± 0.5 | | | |
| $g'_{p,d}$ | 25 ± 8 | | | |

energy values are understood as apparent experimental activation energies $\Delta E'_{p,d}$, $\Delta E'_{n,a}$, to which a temperature dependence according to the observed effective degeneracy factors is superposed (see Appendix B and Eqs. 2.79a, b).

Capture probabilities of the two levels have been determined by Conti and Panchieri [Con71] and Brotherton and Bradley [Bro82a], who former made their measurements near room temperature, the latter around 77 K. The results are depicted in Table 2.4 in form of capture *cross sections* given in the papers. For conversion to capture probabilities the cross sections were multiplied with the following expressions for the mean thermal velocity of electrons respectively holes:

$$v_{th,n} = 2.25 \times 10^7 \sqrt{T/300K} \text{ cm/s}$$

$$v_{th,p} = 1.85 \times 10^7 \sqrt{T/300K} \text{ cm/s}$$

The constants are obtained using in Eq. (2.28) the effective masses $m_n = 0.27\ m_e$, $m_p = 0.4\ m_e$. In the table also the exponents n of the power law temperature dependencies $\sigma_{i,j} \sim T^n$ are shown, as far as determined. As is seen, the results are again very different. Whereas the cross section $\sigma_{p,a}$ according to [Con71] decreases with temperature as $1/T^{4.4}$, in [Bro82a] it is found to be constant. The degeneracy factors of [Con71] differ essentially from the intrinsic values, hence they indicate a significant temperature dependence of the distance of the levels from the respective band.

The injection dependence of the lifetime in n-type silicon calculated with the data of [Con71] shows at low doping a simple monotonous decrease as expected from measurements [Mil76b]. Against this the cross sections of [Bro82a], completed with reasonable degeneracy factors and the temperature dependence of $c_{n,d}$, result in a pronounced intermediate minimum in the $\tau_p(p)$-curves. Since the results of [Con71] are also more complete, they are taken in what follows as a basis.

A series of lifetime measurements concerning the dependence on temperature and doping have been published by Miller and coworkers [Mil82a]. Besides others they observed that the hole lifetime in n-Si with $N_D = 2.6 \times 10^{14}$ cm$^{-3}$ increases from 300 to 400 K by a factor 10 and for a doping concentration of $2.9 \times 10^{15}$ cm$^{-3}$ by a factor 4.3. As function of doping density an inverse proportionality, $\tau_p \sim 1/n_0$, was

found at 400 K, at room temperature the variation with $n_0$ is approximately halved. The inverse proportionality agrees with Eq. (2.85), showing that the condition $n_a, p_d \gg n_0$ must be satisfied at 400 K up to $n_0 = 2.9 \times 10^{15}$ cm$^{-3}$. The data of [Con71] leading at 400 K to $n_a = 5.70 \times 10^{16}$ cm$^{-3}$ and $p_d = 1.23 \times 10^{17}$ cm$^{-3}$ fulfill this condition.

Because these measurements were carried out apparently at low injection levels, they have to be described in the SRH model by the equation

$$\tau_{p,LL} = \frac{1}{N_r c_{p,a}} \frac{(n_0 + n_a)\left(n_0 + \frac{c_{p,d}}{c_{n,d}} p_d\right)}{n_0\left(n_0 + \frac{c_{p,d}}{c_{p,a}} n_a + \frac{c_{p,d}}{c_{n,d}} p_d\right)}, \tag{2.88}$$

which follows for low injection from (2.80). Using the parameter set of [Con71] the calculated temperature dependence is 5–6 times stronger than measured, whereas the doping dependence is nearly as observed. For a better approach to the measurements it is necessary to lower the temperature dependences of $\sigma_{p,a}$ and $\sigma_{n,d}$. This is justified by the constant cross section $\sigma_{p,a}$ of [Bro82a] as well as by measurements on other deep impurities [Han87], which all show significantly smaller absolute n-values for attractive charge states than the values of [Con71]. Taking into account that the high-level lifetime $\tau_{HL}$ will probably increase with temperature, suitable values to reflect the present state of knowledge are $n = -1$ for $\sigma_{p,a}$ and $-2.5$ for $\sigma_{n,d}$. These values are used in the calculations presented below. With this modification the data of [Con71] result in a lifetime increase with temperature which is up to a factor 2.5 stronger than measured, whereas the doping dependence is again approximately reproduced.

Of course the temperature dependence is essentially influenced by the exponential dependencies of $n_a$ and $p_d$ on temperature. For given apparent activation energies, this can be changed via the effective degeneracy factors, see Eqs. (2.79a, b). Without violating the condition for the reciprocal dependency on $n_0$, the value of $g'_{p,d}$ could be lowered by a factor 10. This would enhance however the temperature dependence, because the condition $n_a \gg p_d$ holding then would lead to the approximation (2.81) characterized by the strong T-dependence of $p_d$ and also $c_{n,d}$. With the choice of [Con71] the smaller temperature dependences of $n_a$ and $c_{p,a}$ are effective in addition to $p_d$. Compared with this the approach can be improved somewhat reducing the value of $g'_{n,a}$ to 0.5. With this value the calculated temperature dependence is a factor 1.7 stronger than measured, instead of 2.5 for $g_{n,a'} = 1$. This difference is acceptable considering the significant error sources which lifetime measurements are usually subjected. In the present case possibly a less T-dependent recombination part at the junctions may have influenced the results.

In Fig. 2.21 the injection dependence of the hole lifetime in n-silicon calculated from Eq. (2.80) with the proposed data set is shown for some doping densities and temperatures. The data are depicted. The accordance with the measurements used for adjustment can be checked at the curves indicated with ‘2’ and ‘3’. Between

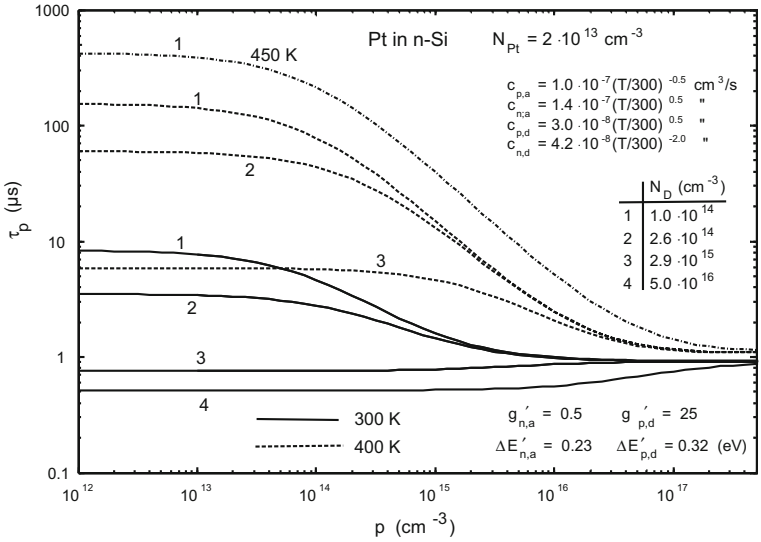



**Fig. 2.21** Minority carrier lifetime for platinum in n-Si as function of injected hole concentration. Parameters are the doping concentration and temperature. The capture rates at 300 K are taken from [Con71]. The temperature dependences of $c_{p,a}$ and $c_{n,d}$ and the degeneracy factor of the acceptor level are reduced compared with [Con71] (see the text)

$1 \times 10^{14}$ and $2.6 \times 10^{14}$ cm$^{-3}$ the low-level lifetime is even at 300 K approximately inversely proportional to the doping concentration, because the condition $n_0 \ll n_a, p_d$ is satisfied in this doping range even at room temperature. At $N_D = 1 \times 10^{14}$ cm$^{-3}$ the lifetime at 300 K decreases with injection level by a factor 9 and at 400 K by a factor 140. From 300 to 400 K the low-level lifetime for $N_D = 1 \times 10^{14}$ cm$^{-3}$ increases by a factor 18.

The disadvantages of the strong decrease of the lifetime with injection level and of the increase with temperature have been pointed out on page 67. The latter property results in a decrease of the forward voltage drop of Pt diffused diodes with increasing temperature [Lut00], an unfavorable property for paralleling devices in modules. An example for the temperature dependence of the forward characteristic of a platinum diffused diode will be given later in Chap. 5 Fig. 5.11. The strong enhancement of lifetime with decreasing injection leads to a somewhat higher recovery time of platinum-diffused devices than in the case of gold. Since especially for thyristors and IGBTs the high low-level lifetime is disadvantageous, platinum is normally not used in fast thyristors and IGBTs.

For diodes these disadvantages associated with the position of the levels are overcompensated for most applications by the small generation current in the space charge region. In steady state the generation rate of the acceptor level is $G_a = e_{n,a}N_{Pt}^- = e_{p,a}N_{Pt}^0$, that of the donor level $G_d = e_{n,d}N_{Pt}^0 = e_{p,d}N_{Pt}^+$. Since $e_{p,a} \ll e_{n,a}$, $e_{n,d} \ll e_{p,d}$, it follows that $N_{Pt}^-, N_{Pt}^+ \ll N_{Pt}^0 \approx N_{Pt}$: platinum in the space charge region is neutral in steady state, similarly as gold. The total generation rate is

$$G = G_a + G_d = \left(e_{p,a} + e_{n,d}\right) N_{Pt}$$
$$= \left(\frac{c_{p,a}}{n_a} + \frac{c_{n,d}}{p_d}\right) n_i^2 N_{Pt}$$

In our dataset with the large $g'_{pd}$ one has $p_d > n_a$, so that the main part of G is contributed by the acceptor level. In agreement with measurements one obtains that the leakage current of Pt diffused diodes is even at 150 °C more than a decade smaller than obtained with gold. Therefore platinum diffusion or particle irradiation is now used in most fast diodes for reduction of carrier lifetime.

In contrast to gold platinum centers are also neutral in weakly doped n base regions in thermal equilibrium because the Fermi level is below the acceptor level. Hence the conductivity is not considerably reduced, an advantage if the diode is integrated together with a unipolar switching element. Furthermore switching of a diode from zero to reverse voltage is not interfered with recharging of platinum ions. At high injection levels the concentrations of Pt in the different charge states are obtained from equations from Appendix B.2 at 300 K as

$$N_{Pt}^{+} : N_{Pt}^{0} : N_{Pt}^{-} = \frac{c_{p,d}}{c_{n,d}} : 1 : \frac{c_{n,a}}{c_{p,a}} = 0.71 : 1 : 1.40$$

All charge states appear in significant fractions, although the negative predominates. The *net* concentration of negative charge, $N_{Pt}^{-} - N_{Pt}^{+}$, amounts to 22% of the total Pt concentration. The platinum charge involved in the switching process from forward to reverse bias is hence of the opposite sign and smaller than in the case of gold. These statements are subjected to the same errors which refer to the capture rates.

**(e) Radiation induced recombination centers**. Often irradiation with high-energy electrons, protons or He ions is used nowadays to generate recombination centers and reduce the carrier lifetime. The energy of irradiation is usually in the range 1 to 15 meV. While electron irradiation produces a homogeneous density of recombination centers, a narrow region of high trap concentration can be created with H and He ion irradiation. The radiation methods show a good reproducibility. As depicted in Fig. 2.22, mainly three independent centers with various levels are generated: the divacancy (VV), the A-center which is an oxygen-vacancy complex (OV), and the

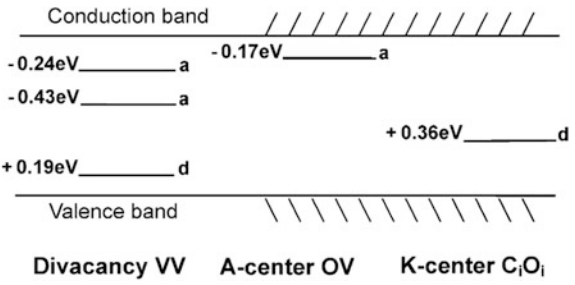



**Fig. 2.22** Energy levels of some important radiation induced centers

K-center, probably an association of an interstitial carbon atom and an interstitial oxygen atom ($C_iO_i$) [Niw08]. The type of the levels and distances from band edges are indicated as in Fig. 2.22. The divacancy has a donor level and two acceptor levels, the upper of which refers to transitions between $R^-$ and $R^{2-}$. In contrast, the A-center has a single acceptor level near the conduction band. The relative concentrations of the centers depend on the radiation energy as well as on tempering processes following the irradiation. For lifetime control in neutral regions, especially at high injection level, the A-center is considered to be most efficient because of high capture rates and high concentrations [Sie02, Sie06]. The carrier generation in space charge regions is determined by the divacancy whose level at 0.43 eV below the conduction band is most effective. Owing to this energy level, the lifetime control by radiation results in a significantly lower blocking current than obtained using gold, but it is higher than generated by platinum. The possibilities to calculate the lifetime behavior in the case of radiation induced centers are still very limited. More details on the radiation technique for lifetime control are given in Sect. 4.9.

## 2.8 Impact Ionization

Impact ionization is a mechanism of carrier generation which takes place at high electric fields and leads to avalanche multiplication. This generation limits the electric field which a blocking junction can sustain without producing a high current, hence it is fundamental for dimensioning of power devices. It determines the breakdown voltage in the whole range above about 6 V in silicon. At the lower end, below about 12 V, where the extension of the high-field region is small enough, quantum mechanical tunneling of carriers becomes significant for the current, and below about 6 V it is decisive for the usable blocking voltage [Hur92a, Hur92b]. Although many power devices contain low-blocking junctions, tunneling is not of high interest for them. The "critical" field strength which leads to avalanche breakdown is a property of the semiconductor. Via the width of the base region required for a given blocking voltage it determines static and dynamic limits of devices. In the present section, we deal with general features of impact ionization, the detailed consequences for power devices will be treated in later sections.

Impact ionization occurs if the electric field is high enough, such that a noticeable number of electrons or holes in the statistical distribution gain sufficient kinetic energy that they can lift a valence electron by impact into the conduction band. Each ionizing carrier generates a pair of a free electron and hole, which again can generate further electron-hole pairs, thus giving rise to an avalanche event. Impact and avalanche generation are therefore used synonymously. Because also the kinetic energy of the secondary particles as following from momentum conservation has to be provided, the ionizing energy which the primary carrier at least

must have is about 3/2 $E_g$ [Moe69]. A few carriers in the thermal distribution have this energy already at zero field strength. However, the generation at zero field is taken into account already as Auger generation, $G_{Aug} = \left(C_{A,n}n + C_{A,p}p\right)n_i^2$, whose microscopic mechanism is identical with that of impact ionization. The latter is defined explicitly as the generation resulting from the *enhancement* of carrier velocities by the field. It is represented by impact ionization rates $\alpha_n$, $\alpha_p$ defined as the number of electron-hole pairs generated per electron or hole per length of the path which the assembly travels with *drift* velocity $v_n$ or $v_p$, respectively. The number of electron-hole pairs generated per unit of time, i.e. the avalanche generation rate $G_{av}$, is then given by

$$G_{av} = \alpha_n \cdot n \cdot |v_n| + \alpha_p \cdot p \cdot |v_p| = \frac{1}{q}\left(\alpha_n \cdot |j_n| + \alpha_p \cdot |j_p|\right) \tag{2.89}$$

On the right-hand side, the densities of the field currents are replaced by the total current densities, the diffusion currents are neglected because of the high fields.

Much experimental and theoretical work has been carried out to determine the ionization rates and their field dependency, which is very strong. Reviews have been given by Mönch [Moe69] and Maes et al. [Mae90]. Theoretically, Wolff predicted very early the relationship $\alpha \propto \exp(-b/\mathsf{E}^2)$ [Wof54]. Experimental ionization rates, on the other hand, were found by Chynoweth [Chy58] and later most other authors, to follow the relationship

$$\alpha = a\, e^{-\frac{b}{\mathsf{E}}}, \tag{2.90}$$

both for electrons and holes. With $\mathsf{E}$ we denote here the positive field strength in direction of the field. In a logarithmic plot versus $1/\mathsf{E}$, the values of $\alpha_n$ and $\alpha_p$ follow

**Table 2.5** Coefficients of ionization rates in Eq. (2.90) as obtained from different publications. T = 300 K

| Source | Electrons | | Holes | | Field for $\alpha_{eff}$ = 100/cm |
|---|---|---|---|---|---|
| Field range ($\mathsf{E}$ in $10^5$ V/cm) | | b ($10^6$ V/cm) | a ($10^6$/cm) | b ($10^6$ V/cm) | ($10^5$ V/cm) |
| Lee et al. [Lee64] $1.8 < \mathsf{E} < 4$ | 3.80 | 1.77 | 9.90 | 2.98 | 1.99 |
| Ogawa [Oga65] $1.1 < \mathsf{E} < 2.5$ | 0.75[a] | 1.39[a] | 0.0188[a] | 1.54[a] | 1.88[a] |
| | | | 4.65[b] | 2.30[b] | 1.80[b] |
| Overstraaten, De Man [Ove70] $1.75 < \mathsf{E} < 4$ | 0.703 | 1.231 | 1.582 | 2.036 | 1.66 |
| Our choice [Sco91] $1.5 < \mathsf{E} < 4$ | 1.10 | 1.46 | 2.10 | 2.20 | 1.81 |

[a]directly measured [b]from $\alpha_{eff}$, $\alpha_n$

a straight line. Shockley derived this relationship [Sho61] using a simplified model for the manner the carriers reach the high ionization energy. As shown then by Baraff [Baf62], the field dependences of Wolff and of Chynoweth are limiting cases of a more general theory. The experimental determination is done usually by measuring carrier multiplication factors $M_n$ and $M_p$, consisting of double integrals over the ionization rates (see Appendix C), which then have to be extracted. The measured field dependence obeys the relationship (2.90) down to low fields, as is not expected to this extent from theory [Mae90, Oga65]. Table 2.5 contains the constants a and b for silicon at room temperature as gathered from some often cited papers. From Ogawa's work [Oga65], two sets for $\alpha_p$ are given, the first one determined from multiplication factors like other values in the table. The second set describes the hole ionization rate determined in [Oga65] from measurements of the electron ionization rate $\alpha_n$ and the *effective* ionization rate

$$\alpha_{eff} = \frac{\alpha_n - \alpha_p}{\ln(\alpha_n/\alpha_p)} \tag{2.91}$$

which was determined from the breakdown voltage of pin diodes (see below). The extremely varying values of a (= $\alpha(\mathsf{E} = \infty)$) in the table are mainly due to the extrapolation with different slopes in the plot versus $1/\mathsf{E}$ owing to the different b-values. Nevertheless, also in the experimental range the diversity of the obtained ionization rates is large. For example, the $\alpha_{eff}$– values obtained from [Lee64] and [Ove70] at a field strength of $2 \times 10^5$ V/cm differ by more than a factor 4. Although the differences in the *field* for a given $\alpha_{eff}$ are smaller because of the strong field dependence, they too are considerable. In the last column of Table 2.3, the field strengths belonging to $\alpha_{eff} = 100$/cm are given. This field can be interpreted as the critical field $\mathsf{E}_c$ of a pin diode whose width of the intrinsic i-region is $w = 1/\alpha_{eff} = 100$ µm, since the breakdown condition of a pin diode is $\alpha_{eff}(\mathsf{E}_c) \cdot w = 1$ (see Eq. (2.93) below and Chaps. 3 and 5 on pn-junctions and pin-diodes). The breakdown voltage $V_B = w \cdot \mathsf{E}_c$ calculated with the data of [Lee64] is 20% higher than obtained from [Ove70]. For $p^+$n junctions the difference is still larger (about 35% for $V_B > 1000$ V).

Calculations for a tight dimensioning of devices need ionization rates whose results are in close agreement with the measured blocking behavior. We have compared calculations using different α-sets with measurements on thyristors and diodes, whose weakly doped n base region was very homogeneously doped by neutron transmutation (see the chapter on technology), a method not used before 1974. Mainly devices in the blocking range 1200 to 6000 V were used for comparison, although data in the lower blocking range have also been considered. The ionization rates of Ogawa with the second set of $\alpha_p$ is found to be well suited for *diodes* with blocking voltage > 300 V. The blocking behavior of pn$^-$p-structures in thyristors, however, was found to be somewhat better described by a course of the hole ionization rate *between* the two strongly differing $\alpha_p$-curves of [Oga65]. From these comparisons and by considering also measurements with low-blocking devices [Lee64, Ove70], the α parameters given in the last line of Table 2.5 are

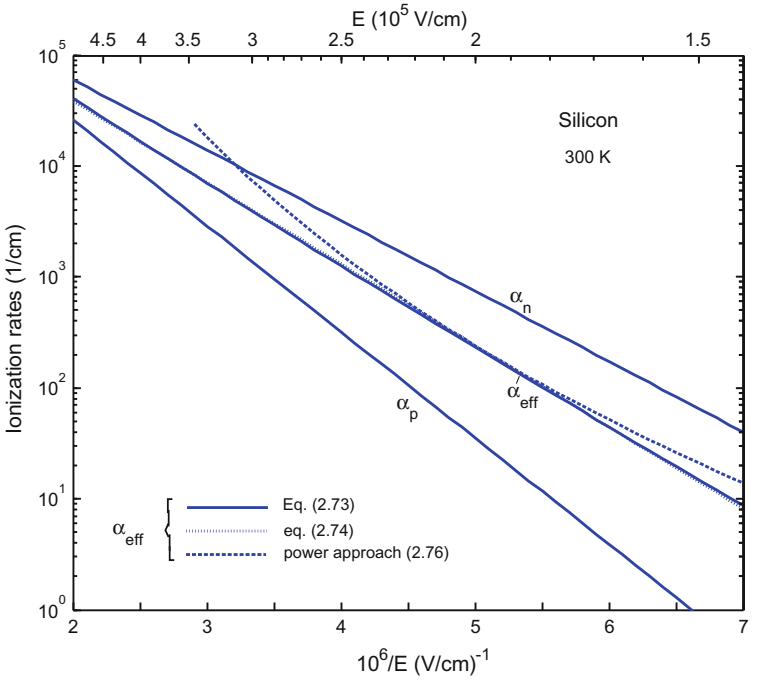



**Fig. 2.23** Field dependence of ionization rates in silicon. The power approach (2.94) [Ful67, Shi59] is matched at the point $\mathsf{E}_0 = 2 \times 10^5\,\mathrm{V/cm}$ using Eq. (2.95)

proposed for the simulation of power devices [Sco91]. They agree also well with a renewed determination of the ionization rates reported in a short paper of Valdinoci et al. [Val99], although these authors use a more complex fitting expression.

In Fig. 2.23 these ionization rates for silicon at 300 K (Table 2.5, last line) are plotted versus the inverse field strength. Within a relative small field range the α's vary by several orders of magnitude. The ionization rate of electrons is much larger than that for holes, and the effective ionization rate defined by (2.91) runs between $\alpha_n(\mathsf{E})$ and $\alpha_p(\mathsf{E})$. As is seen, $\alpha_{eff}$ too can be described with excellent approximation by a straight line in this plot and hence can be described by (2.90). Best agreement with (2.91) together with the parameter set in the last line of Table 2.5 is obtained with

$$\alpha_{eff} \approx 1.06 \times 10^6 \cdot e^{-1.68 \cdot 10^6/E} = a_{eff} \cdot e^{-b_{eff}/\mathsf{E}} \tag{2.92}$$

which up to a few percent agrees with the exact $\alpha_{eff}$ in the range $1.5 \times 10^5$ to $4 \times 10^5$ V/cm. Equation (2.92) differs only very little from the result of Ogawa [Oga65], who obtained

$$\alpha_{eff} = 1 \times 10^6 \exp(-1.66 \cdot 10^6/\mathsf{E})/\text{cm}.$$

The effective ionization rate (2.91) is defined in a manner that solely this $\alpha_{eff}$ determines the breakdown voltage of pn-junctions [Wul60, Oga65]. The breakdown condition which the field $\mathsf{E}(x)$ and hence voltage at breakdown must satisfy is

$$\int \alpha_{eff}(\mathsf{E}(x))dx = 1 \tag{2.93}$$

where the integration extends over the space charge region. The derivation is given in Appendix C. Although this holds exactly only if the ratio $\alpha_n/\alpha_p$ is independent of $\mathsf{E}$, it is a very good approximation also in most other cases (see Appendix C). To enable easy analytical integration for simple field shapes, Shields [Shi59] and Fulop [Ful67] approximated the field dependency by a power law, which normalized to a field $\mathsf{E}_0$ reads

$$\alpha_{eff}(\mathsf{E}) \cong C\left(\frac{\mathsf{E}}{\mathsf{E}_0}\right)^n \tag{2.94}$$

Based on this approach, practicable and often used relationships for the breakdown voltage can be derived (see chaps. 3 and 5). If (2.94) together with its derivative is fitted to (2.92) at the point $\mathsf{E}_0$, the constants are obtained as

$$n = b_{eff}/\mathsf{E}_0 \qquad C = a_{eff} \cdot \exp(-n) \tag{2.95}$$

where according to (2.92) $a_{eff} = 1.06 \times 10^6$ cm$^{-1}$ and $b_{eff} = 1.68 \times 10^6$ V/cm. The value n = 7 used by Shields and Fulop is obtained for $\mathsf{E}_0 = b_{eff}/7 = 2.40 \times 10^5$ V/cm. Matching at the field $\mathsf{E}_0 = 2 \times 10^5$ V/cm, Eq. (2.95) yields $n = 8.40$ and $C = 238$ cm$^{-1}$. Using these values, approximation (2.94) is plotted additionally in Fig. 2.20. Although the approximation is not very good over a wide range of $\mathsf{E}$, very satisfying results can be obtained choosing the matching point $\mathsf{E}_0$ near the maximum of the considered field distribution.

Whereas for diodes only $\alpha_{eff}$ is decisive for the breakdown voltage, for transistor structures such as the pn$^-$p structure in thyristors and IGBTs both ionization rates separately have an effect on the breakdown behavior. Of course, also the single ionization rates $\alpha_n$ and $\alpha_p$ can be approximated by a power law according to (2.94), (2.95). In this book, the Shields-Fulop approach will often be used to describe avalanche multiplication and blocking capability in an analytical way. If the *field* distribution $\mathsf{E}(x)$ in (2.93) does not allow an analytical integration, the power approximation loses its meaning.

The ionization rates decrease with increasing temperature because the mean free path between collisions with phonons decreases. As shown by Grant [Gra73] and Maes et al. [Mae90], the temperature dependence can be expressed by an increasing coefficient $b$ in (2.90), while the pre-exponential factor $a$ may be left constant. For electrons, the temperature coefficient $db/dT$ was determined in [Gra73] to be

1300 V/(cmK), whereas from [Mae90] a mean value of 710 V/(cmK) is obtained; for holes Grant found $db/dT$ = 1100 V/(cmK). Using a value of 1100 V/(cmK) in both cases, the temperature dependence of blocking behavior in the range −20 to 150 °C is well described according to our observations. Hence the following field and temperature dependences are found to be well suited for power devices:

$$\begin{aligned} \alpha_n &= 1.1 \times 10^6 \cdot e^{-\frac{1.46\times10^6 + 1100(T-300K)}{\mathsf{E}}}\ \mathrm{cm}^{-1} \\ \alpha_p &= 2.1 \times 10^6 \cdot e^{-\frac{2.2\times10^6 + 1100(T-300K)}{\mathsf{E}}}\ \mathrm{cm}^{-1} \end{aligned} \tag{2.96}$$

where the field is scaled in V/cm. Inserting this into (2.91) and using (2.92) at 300 K, the same T-dependence is obtained for $\alpha_{eff}$:

$$\alpha_{eff} \cong 1.06 \times 10^6\ e^{-\frac{1.68\times10^6 + 1100\cdot(T-300\ K)}{\mathsf{E}}}\ \mathrm{cm}^{-1} = a_{eff}\, e^{-\frac{b_{eff}(T)}{\mathsf{E}}} \tag{2.97}$$

The temperature dependence of the constants in the Shields approximation follows inserting the coefficient $b_{eff}(T)$ of the last equation into (2.95):

$$\begin{aligned} n(T) &= \frac{b_{eff}(T)}{\mathsf{E}_0} = \frac{1.68 \times 10^6\ + 1100 \cdot (T - 300\ K)}{\mathsf{E}_0} \\ C(T) &= \frac{a_{eff}}{e^{n(T)}} = \frac{1.68 \times 10^6\ /\mathrm{cm}}{e^{n(T)}} \end{aligned} \tag{2.98}$$

In the mentioned paper of Valdinoci et al. [Val99], the temperature dependency of the ionization rates in the range 300 – 670 K is described. The results [Val99] agree well with the above Eqs. (2.96). Singh and Baliga [Sin93] have determined the temperature dependence of the effective ionization rate in the range 77 – 300 K using the approach of Shields and Fulop. Although the temperature dependences of their constants differ partly strongly from (2.98), the ionization rate $\alpha_{eff}$ itself varies quite similarly (by a factor 2.4 from 300 to 100 K).

As mentioned in Sect. 2.1, SiC has the advantage of a very high critical field. Measurements of the ionization rates in 4H-SiC, the preferred polytype, have been performed by Konstantinov et al. [Kon98], Ng et al. [Ng03] and Loh et al. [Loh08]. The results were fitted by the following equations [Loh08], a modified form of (2.90):

$$\alpha_n = 2.78 \times 10^6 \exp\left[-\left(\frac{1.05 \times 10^7}{\mathsf{E}}\right)^{1.37}\right] \mathrm{cm}^{-1} \tag{2.99}$$

$$\alpha_p = 3.51 \times 10^6 \exp\left[-\left(\frac{1.03 \times 10^7}{\mathsf{E}}\right)^{1.09}\right] \mathrm{cm}^{-1} \tag{2.100}$$

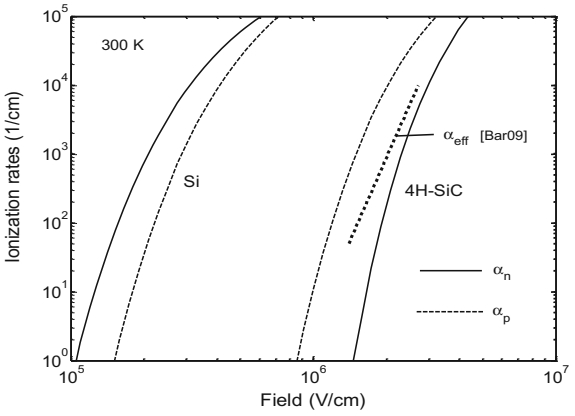



**Fig. 2.24** Ionization rates of 4H-SiC and Si at 300 K. SiC-curves after [Kon98, Loh08, Ng03] ($\alpha_n, \alpha_p$) and [Bar09] ($\alpha_{eff}$)

where E has to be used in V/cm. These ionization rates together with those of silicon (2.96) are plotted in Fig. 2.24 versus E. As is seen, the field required for a given value of ionization rates is in 4H-SiC nearly an order of magnitude higher than in silicon. Contrary to silicon, the hole ionization rate $\alpha_p$ is larger than $\alpha_n$ in 4H-SiC.

An independent determination of the *effective* ionization rate has been published by Bartsch et al. [Bar09], who use measurements of the breakdown voltage of 4H-SiC $p^+n^-n^+$-diodes with different doping concentrations of the base region. From their measurements, which include temperature dependence, they obtain that

$$\alpha_{eff} \cong 2.18 \cdot 10^{-48}\, E^{8.03} \quad \mathrm{cm}^{-1} \ (\mathsf{E}\ \mathrm{in\ V/cm}) \tag{2.101}$$

at 300 K. This dependency is plotted in Fig 2.24 additionally. The agreement with the other SiC curves on the whole is good. Nevertheless, considering (2.99), (2.100) and (2.91) on one side and (2.101) on the other the difference in the results is practically not unimportant. In the first case the effective α in the surrounding of $10^3$/cm can be written in the power approach as $\alpha_{eff} \approx 3.43 \cdot 10^{-54} E^{9.02} cm^{-1}$. According to this, the field for the ionization rate 1000/cm is 1.82 MV/cm. According to Eq. (2.101) however the same ionization rate is reached only at 2.04 MV/cm. For a triangular field shape this means that the breakdown voltage according to (2.101) [Bar09] is 25% higher for the same doping concentration than according to (2.99), (2.100). This is probably the result of improvements in material quality of the SiC wafers and in SiC device manufacturing processes since the earlier measurements.

## 2.9 Basic Equations of Semiconductor Devices

The device operation depends on the processes with which the carriers and the electric field in the interior react on terminal currents and voltages. This is described by some basic equations which we will discuss now. A central part of these equations are the continuity equations of electrons and holes

$$-\frac{\partial n}{\partial t} = \; div\, \vec{\mathsf{J}}_{\mathsf{n}} + \; R_n \tag{2.102}$$

$$-\frac{\partial p}{\partial t} = div\, \vec{\mathsf{J}}_{\mathsf{p}} + R_p \tag{2.103}$$

Here $\vec{J}_n, \vec{J}_p$ are the vectors of particle current densities. The equations represent the time decrease of a carrier concentration (e.g. $-\partial n/\partial t$) as a flow of carriers out of the considered volume element ($div\, \vec{\mathsf{J}}_{\mathsf{n,p}}$) plus a disappearance of carriers with a rate $R_n$ respectively $R_p$. In these exact mathematical equations one has to insert the previously derived physical relationships and models for the current densities and excess recombination rates $R_n$, $R_p$. The latter have to include generally, besides the thermal recombination-generation rates treated in Sect. 2.7 and called now $R_{n,p,th}$, the impact generation rate $G_{av}$: $R_{n,p} = R_{n,p,th} - G_{av}$. The electrical current densities have been given in one-dimensional form in (2.43a, 2.43b), (2.44). In two or three dimensions the particle current densities are

$$\vec{\mathsf{J}}_n = -\mu_n n\, \vec{\mathsf{E}} - D_n\, grad\; n \tag{2.104}$$

$$\vec{\mathsf{J}}_p = \mu_p p\, \vec{\mathsf{E}} - D_p\, grad\, p \tag{2.105}$$

The continuity equations include the law of conservation of charge

$$div\, \vec{j} + \frac{\partial \rho}{\partial t} = 0 \tag{2.106}$$

where $\vec{j}$ is the total electrical current density transported by the carriers (conduction current), $\vec{j} = q(\vec{J}_p - \vec{J}_n)$, and $\rho$ the charge density or 'space charge'. Equation (2.106) is obtained by subtracting (2.102) from (2.103) and considering that the difference of the excess recombination rates $R_n - R_p$, if it is non-zero, signifies a recharging of impurities, usually of the recombination centers. Hence $R_n - R_p$ contributes to a change of charge density, which contains the charged flat and deep impurities in the form:

$$\rho = q(p - n + N_D^+ \; - N_A^- + N_r^+ \; - N_r^-) \tag{2.107}$$

A third differential equation on the level of the continuity equations results from the fundamental law

$$div\ \vec{D} = \rho \tag{2.108}$$

which states that an electric charge is a source of a displacement field $\vec{D}$. The latter is proportional to the electric field strength, $\vec{D} = \varepsilon\vec{\mathsf{E}}$, where the permittivity constant $\varepsilon$ splits up into the absolute permittivity (permittivity of vacuum) and the relative permittivity $\varepsilon_r$ as a material constant, $\varepsilon = \varepsilon_r \cdot \varepsilon_0$. The $\varepsilon_r$-values of some materials are compiled in Appendix D. With (2.107) Eq. (2.108) can be written as follows:

$$div\ \varepsilon\vec{\mathsf{E}} = \rho = q(p - n + N_D^+ - N_A^- + N_r^+ - N_r^-) \tag{2.109}$$

The further procedure now is to express the electric field in (2.104), (2.105) and (2.109) as the negative gradient of a potential $V$:

$$\vec{\mathsf{E}} = -grad\ V \tag{2.110}$$

Equation (2.109) then turns into the Poisson equation:

$$div\ grad\ V = -\frac{q}{\varepsilon}(p - n + N_D^+ - N_A^- + N_r^+ - N_r^-) \tag{2.111}$$

where the semiconductor is assumed homogeneous and isotropic with regard to $\varepsilon$. For non-cubic crystals $\varepsilon_r$ is a tensor, but in current simulation programs it is assumed to be a scalar. The differential operator on the left hand side of (2.111) is $div\ grad = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$, the Laplace operator.

The Poisson equation and the continuity Eqs. (2.102), (2.103) with (2.110) substituted for $\vec{\mathsf{E}}$ in the current densities form a system of three partial differential equations with the unknown variables $V$, $n$ and $p$. The doping structure and the density of recombination centers as well as boundary conditions at the surface and the characteristics of the external circuit are required to be known. The normal donors and acceptors are assumed usually to be completely ionized and the concentration of deep impurities to be small. The concentration of charged impurities is then given by the net doping concentration $N_D - N_A$, and the recombination rates $R_n$, $R_p$ are approximately equal (see Sect. 2.7). The three differential equations are then sufficient to calculate the potential $V$ and the carrier concentrations n and p as functions of space and time. The Eqs. (2.102)–(2.103) and (2.110), (2.111) with $N_D^+ = N_D$, $N_A^- = N_A$, $N_r^+ = N_r^- = 0$ and $R_n = R_p = R$ are called therefore the basic semiconductor equations. The behavior of this system is investigated in the book of Selberherr [Sel84]. Simulation programs based on these equations in one to three dimensions are on the market. The two- and three-dimensional programs are rather complex with all implications. In special one-dimensional cases the equations are used widely as starting point for analytical calculations.

In fast switching devices, the charge of deep traps is often considerable, and additionally the incomplete ionization of normal dopants can be a subject of interest, particularly in semiconductors like SiC. In these cases, (2.109), (2.110) and (2.111) have to be used in their general form, and for each impurity level, whose variable partial occupation has to be considered, one has to add an equation of the form (2.78) to the system. A program including these effects is Sentaurus$^{TCAD}$ [SYN07].

So far we have assumed that the device has the same temperature at all points. Often this is a good approximation because the thickness of the semiconductor chips is relatively small and the thermal conductivity of silicon is high. On the other hand, phenomena of current constriction, thermal run-away and other processes responsible for destruction at heavy loads are connected often with a strongly inhomogeneous temperature distribution. For power devices, heat conduction is the primary criterion to be considered for the area and package of a device. To supplement the above system for inclusion of variable temperature and to provide a basis for thermal estimates, one has to add the equation of heat conduction. The heat current density, i.e. the heat energy transported through a surface element per unit area and time, is proportional to the negative gradient of temperature

$$J_{heat} = -\lambda \cdot grad\ T \tag{2.112}$$

where the proportionality factor λ is the thermal conductivity. The thermal energy per volume belonging to a temperature rise $\Delta T$ is $Q = \rho_m \cdot c \cdot \Delta T$, where $\rho_m$ denotes the specific (mass) density and c the specific heat of the material. Using this, the continuity equation describing the conservation of thermal energy is obtained in the form

$$\rho_m c \cdot \frac{\partial T}{\partial t} = div\ (\lambda \cdot\ grad\ T) + H \tag{2.113}$$

where $H$ signifies the heat generation rate per unit volume. This consists of the ohmic energy dissipation $\vec{\mathsf{E}} \cdot \vec{j}$ and the heat produced by the net recombination of carriers, $R \cdot E_g$ [Sel84]:

$$H = \vec{\mathsf{E}} \cdot \vec{\mathsf{j}} + R \cdot E_g \tag{2.114}$$

where the non-degenerate case is assumed. $R$ is defined here again as the total net recombination rate including the impact generation: $R = R_{th} - G_{av}$. By adding the heat flow equation Eq. (2.113) to the above system of partial differential equations, the temperature distribution can be calculated in addition to $V$, $n$ and $p$. All quantities in these equations, such as the mobilities and the carrier lifetime, have to be used in their temperature dependent form. The equations of current densities may be unchanged for a first approximation only. A more detailed discussion can be found in [Sel84]. We note that in silicon $\rho_m \cdot c$ is nearly independent of temperature and has the value $\rho_m \cdot c = 2.0\ Ws/(\mathrm{cm}^3\mathrm{K})$, whereas the heat conductivity $\lambda$ decreases

considerably with T. In the range 220–600 K, the temperature dependence of λ is described by the formula [Kos74]:

$$\lambda = \frac{320}{T - 82} \quad \frac{W}{\mathrm{cmK}} \tag{2.115}$$

where $T$ is scaled in Kelvin. The one-dimensional increase of temperature from the lower surface of a chip towards the interior can be simply estimated for a given load using these equations.

Till now the ***Maxwell equations*** as the general description of all electro-magnetic phenomena have not been used. The question arises, how far they are satisfied by the above calculation method. Law (2.108) is one of the Maxwell equations. Furthermore, the equation of conservation of charge (2.106) can be deduced from the first Maxwell equation

$$rot\ \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t} \tag{2.116}$$

by applying the div operator und using (2.108). Beyond this, however, the Maxwell equations are not taken into account. According to (2.116) the current is accompanied with a magnetic field $\vec{H}$ and a magnetic induction $\vec{B} = \mu\mu_0\vec{H}$. Fast changes of the current density $\vec{j}$ and the displacement current density $\partial\vec{D}/\partial t$ cause a corresponding variation of $\vec{H}$ and $\vec{B}$, and according to the second Maxwell equation

$$rot\ \vec{E} = -\frac{\partial \vec{B}}{\partial t} \tag{2.117}$$

this leads to an induced electric field. Since $rot\ \vec{E} \neq 0$, this field cannot be represented by the gradient of a potential as in (2.110); hence it is not contained in the described differential equations. Effects of self-induction and current crowding (skin effect), which play an important part in metal wires, are therefore ignored. Due to the smaller current densities in semiconductors these effects are significantly smaller. In the current equations above we have omitted furthermore the Lorentz force $\pm\ q\ \vec{v} \times \vec{B}$ on the carriers. External magnetic fields as used for the Hall effect (see Sect. 2.4) are excluded therefore. Apart from this, the omission of the Lorentz force is in line with disregarding the second Maxwell equation, both neglecting the feedback of the induced magnetic field on the current distribution. Switching processes where these effects become significant have not been demonstrated however till now to our knowledge. For the great majority of possible applications the discussed basic device equations form an adequate basis for a realistic device simulation.

## 2.10 Simple Conclusions

We consider now a few simple but important consequences of the above equations. A homogeneous n-type semiconductor is assumed, and on one hand the decay of a small minority carrier density, on the other hand the decay of a small charge density is studied, both as a function of time after external generation and as function of the distance x from the surface in the case of a stationary generation at the surface.

### *2.10.1 Temporal and Spatial Decay of a Minority Carrier Concentration*

A low injection hole density $\Delta p = p$ in a neutral n region is assumed, the hole charge is neutralized then by an equal excess electron concentration $\Delta n \equiv n - n_0 = \Delta p$.

**(a) Temporal decay of a homogeneous hole concentration**: This decay is obtained from (2.103) with $div\,\vec{J}_p = 0$ and has been given already by Eq. (2.50). The minority concentration decays with the time constant $\tau_p$, the minority carrier lifetime.

**(b) Spatial decay for a stationary generation at the surface:** Since the current density $j_n + j_p = 0$, it follows from (2.43a), (2.43b) that a so-called "Dember field"

$$\mathsf{E} = -\frac{D_n - D_p}{\mu_n n}\frac{dp}{dx}$$

is necessary to maintain this condition. Since $p \ll n$, the hole current caused by this field is however small against the diffusion current, hence $J_p = -D_p dp/dx$. Using this in (2.103) together with $\partial p/\partial t = 0$ and $R_p = \Delta p/\tau_p$ one obtains

$$-D_p\frac{d^2p}{dx^2} + \frac{p}{\tau_p} = 0$$

The solution is

$$p(x) = p(0)\cdot e^{-x/L_p}$$

with

$$L_p = \sqrt{D_p\tau_p} \tag{2.118}$$

This decay length by which the minority carrier concentration decreases is called **diffusion length** of the minority carriers (here holes), indicating that the spreading

takes place by diffusion. $L_p$ can be adjusted by the minority carrier lifetime $\tau_p$. With $D_p = kT/q \cdot \mu_p = 12$ cm$^2$/s (for small doping density) a lifetime of 1 µs results in a diffusion length of 34.6 µm. The diffusion length of electrons is somewhat larger for a given lifetime because of the higher diffusion constant $D_n$. The diffusion lengths in devices are chosen often with reference to the base width $w_B$. With the appropriate lifetime, diffusion lengths of several hundred microns are possible. Compared with a charge density considered in next section under b) the minority carriers gain a much larger distance from the point of generation.

### 2.10.2 Temporal and Spatial Decay of a Charge Density

A small charge density or space charge $\rho = -q \cdot \delta n$ with $\delta n << n_0$ is assumed, the deviation $\delta n$ from the equilibrium $n_0$ being not neutralized by an equal hole concentration.

**(a) Temporal decay of a homogeneous charge density**: From the equation of conservation of charge (2.106) one obtains with $j = q \cdot \mu_n \cdot n_0 \cdot \mathsf{E}$ and $div\ \mathsf{E} = \rho/\varepsilon$

$$\frac{q \cdot \mu_n \cdot n_0}{\varepsilon} \rho + \frac{dp}{dt} = 0$$

This results in the time decay

$$\rho(t) = \rho(0) \cdot e^{-t/\tau_{rel}} \tag{2.119}$$

where the time constant is the **relaxation time**

$$\tau_{rel} = \frac{\varepsilon}{q \cdot \mu_n \cdot n_0} = \frac{\varepsilon_r \varepsilon_0}{\sigma} \tag{2.120}$$

$\tau_{rel}$ is inversely proportional to the electrical conductivity $\sigma$ and very small; in silicon at $n_0 = 1 \times 10^{15}$ cm$^{-3}$ ($\mu_n = 1350$ cm$^2$/Vs, $\sigma = 0.22$ A/Vcm) one obtains $\tau_{rel} = 4.8$ ps. A homogeneous space charge left to itself can exist only very shortly.

**(b) Decrease of a space charge with distance x from the surface**: The stationary decrease towards the inner is considered starting from a constant charge density $\rho(0) = -q \cdot \delta n(0)$ at the surface, caused for example by a positive voltage at an isolated gate electrode. Due to the nearly zero hole concentration the hole current is negligible, hence the total current density $j = j_n$. Since $d\rho/dt = 0$, one obtains from the equation of charge conservation (2.106) using (2.43a) together with the Einstein relation (2.44)

$$0 = \frac{dj}{dx} = q\mu_n \left( n\frac{d\mathsf{E}}{dx} + \frac{dn}{dx}\mathsf{E} + \frac{kT}{q}\frac{d^2n}{dx^2} \right)$$

The term $dn/dx \cdot \mathsf{E}$ as a product of two small quantities is negligible. Hence, using (2.108) and $\delta n = -\rho/q$, (2.118) turns into

$$\frac{n_0}{\varepsilon}\,\rho \;-\; \frac{kT}{q^2}\,\frac{d^2\rho}{dx^2} \;=\; 0$$

This yields again an exponential decay with distance x:

$$\rho(x) = \rho(0) \cdot e^{-\frac{x}{L_D}} \tag{2.121}$$

where now

$$L_D = \sqrt{\frac{\varepsilon_r \varepsilon_0 \cdot kT}{n_0 \cdot q^2}} = \sqrt{D_n \tau_{rel}} = 0.409 \sqrt{\frac{T/300\,K}{n_0/10^{14}\mathrm{cm}^{-3}}}\ \mu\mathrm{m} \tag{2.122}$$

This length determining the spatial decay of a charge density is called **Debye length**. It is inversely proportional to the equilibrium carrier density. The numerical expression on the right-hand side is obtained for silicon with $\varepsilon_r = 11.7$. Even at $n_0 = 1 \times 10^{14}$ cm$^{-3}$, $L_D$ amounts only to 0.41 µm at 300 K. Hence the space charge decreases very rapidly towards zero. The Debye length is typically two or three orders of magnitude smaller than the diffusion lengths. The relation between $L_D$ and the relaxation time is similar to the relation between the minority carrier diffusion length and lifetime. The *majority* carrier concentration and mobilty appearing in the Debye length (2.122) and the relaxation time (2.120) show that the spreading and decay of the space charge is a majority carrier effect.

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